Week 6 — Hypothesis testing

Business Forecasting

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My code (.R) Notes

Where we are

Describing data → Reasoning under uncertainty → Regression → Time series forecasting

Reasoning under uncertainty. Last week: a range around an estimate. This week: testing a claim about a parameter, and what a p-value does and does not tell you.

- This week (Testing I): one-sample hypothesis testing — hypotheses, test statistic, rejection region, Type 1 & 2 errors, power, and p-values.

1 · What a hypothesis test is

A claim about a population, and a rule for deciding whether one sample argues against it.

Hypothesis Testing

Can we get a higher price if cleanliness level is above 4.5?

This is a hypothesis testing question: we see which of two competing hypotheses is more likely given the data.

- **Hypothesis** is a claim about a parameter or a distribution.

1. H_0 Null Hypothesis: the claim to be tested, the skeptical perspective.

- Prices of clean and dirty apartments are equal
- $H_0 : \mu_c = \mu_d$

2. H_A Alternative Hypothesis: an alternative claim under consideration.

- Prices of clean apartments are higher
- $H_A : \mu_c > \mu_d$

- Assume the null is true and see if the data reject it in favor of the alternative.
- You either reject the null, or fail to reject it (never “accept” it).

Hypothesis testing: more examples

Some examples of hypotheses:

- You released a promotion (10% off) to some customers. Do they spend a different amount than those who didn't get it?

- $H_0 : \mu_p = \mu_n$ (promotion group spends the same)
- $H_A : \mu_p \neq \mu_n$ (promotion group spends a different amount)

- Your courier's contract promises an average delivery time of 100 minutes. You can claim a discount only if you prove they are slower. Is the true mean delivery time above 100 minutes?

- $H_0 : \mu = 100$ (the courier delivers as promised)
- $H_A : \mu > 100$ (the courier is slower than promised)

Hypothesis testing: why a test at all?

- Suppose we time our sample of 36 deliveries and get a mean of 103 minutes. That is above the promised 100.
- Why would we need a fancy test? Why not claim the discount right away?
- Because we only have a sample, but we want to learn about the population parameter!
- Maybe those 36 deliveries averaged 103 minutes, while across all the courier's deliveries the true mean is 100 or less.
- That is the whole job of a test: decide whether the gap we see is a real difference in the population, or just sampling noise.
- Hypotheses are always about parameters! Never about sample statistics.

Exercise 1 — State the hypotheses

A report claims the true mean monthly electricity bill of Mexican households is 200. We want to test whether it differs from 200. Which pair states the hypotheses correctly?

- $H_0 : \mu = 200$ vs $H_A : \mu \neq 200$
- H_0: sample mean $= 200$ vs H_A: sample mean $\neq 200$
- $H_0 : \mu \neq 200$ vs $H_A : \mu = 200$
- $H_0 : \mu > 200$ vs $H_A : \mu < 200$

$H_0 : \mu = 200$ **vs** $H_A : \mu \neq 200.$

Hypotheses are always about the population parameter μ, never the sample mean. “Differs from” is two-sided ($\neq$), and the null is the claim being tested ($\mu = 200$).

Testing Procedure Summary

1. Set up your hypotheses
2. Calculate your test statistic
3. Check it against the rejection region
4. Decision: Reject or fail to reject the null hypothesis

We will now go into the details of each step.

Testing Procedure

A test procedure is a rule based on sample data for deciding whether to reject the null.

It relies on two elements:

- **Test statistic**
  - A function of the sample data that summarizes evidence for or against the hypothesis
  - We know how it is distributed under the null (e.g. standardized mean)
- **Rejection region**
  - Values of the test statistic for which we reject the null
    - These are values that are very unlikely under the null hypothesis
  - Typically threshold-based: reject H_0 if the test statistic > k

2 · Where the decision rule comes from

One courier, one sample, and the distribution of results you would see if the claim were true.

Testing Procedure: Example

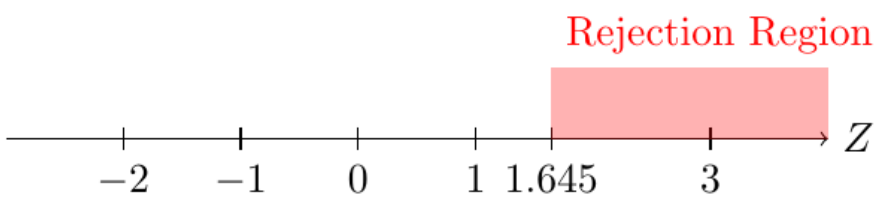
The courier contract again: you can claim the discount only by showing the mean delivery time is above 100 minutes. No evidence, no discount — so "above 100" is the claim that has to be proven, which makes it the alternative.

You time $n = 36$ deliveries. From years of records the standard deviation is known, $\sigma_x = 20$ minutes.

- **Test statistic**:

$$z = \frac{\bar{x} - 100}{\underbrace{\sigma_x / \sqrt{36}}_{SE}}$$

- **Rejection region**: for a 5% significance level, reject if $z \in \{1.645, \infty\}$



The claim we will test

Our running data: `deliveries` — the courier's last 36 logged deliveries. One row per delivery; `minutes` is the door-to-door delivery time.

The contract claims the true mean delivery time is 100 minutes.

Before any test: where is the sample centred, and how variable is it?

Exercise 2 — Describe the sample

Using `deliveries$minutes`, over the $n = 36$ deliveries:

1. Look at the first few rows, so you know what one observation is.
2. What is the sample mean, and the sample standard deviation?
3. What is the standard error of the mean?

Exercise | Start Over | Show Hint | Run Code

```
1 mins <- deliveries$minutes
```

Exercise 3 — The test statistic by hand

Write the test statistic for $H_0 : \mu = 100$ against $H_A : \mu > 100$ yourself, from the definition $\dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$.

Exercise | Start Over | Show Hint | Run Code

```
1  mins <- deliveries$minutes
2  n    <- 36
```

Is 2.4 far?

You know the sample mean sits 2.4 standard errors above 100. Is that far? Nothing so far tells us.

To judge 2.4 we need something to compare it with.

So ask: if the courier really did deliver in 100 minutes on average, what numbers would we get? Some samples of 36 deliveries would land above 100 by luck, some below. Each sample gives a statistic.

Do that many times and you can see which values are ordinary and which are rare. Then look where our 2.4 falls.

🎲 Draw the world where the courier keeps the promise

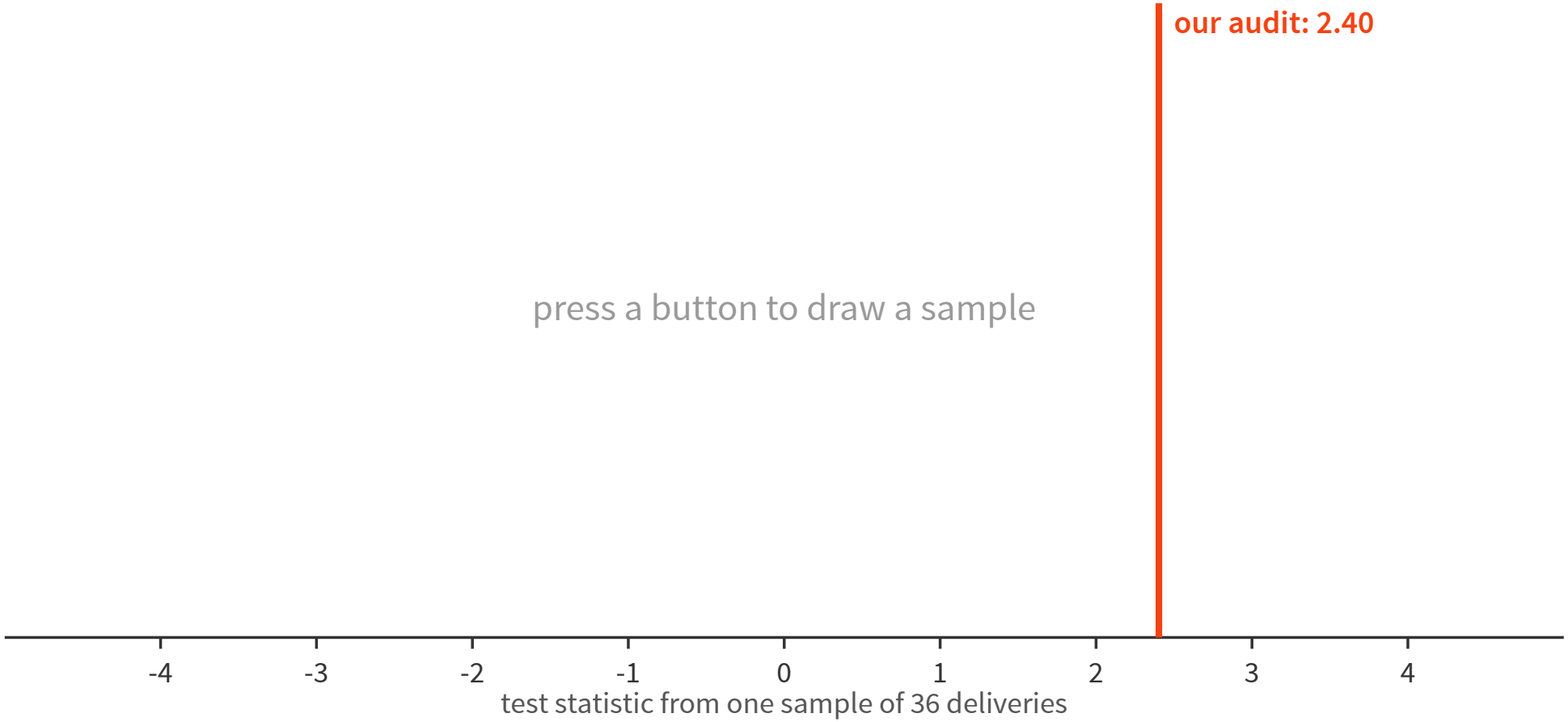
Every draw is one new sample of 36 deliveries, from a courier whose true mean really is 100 minutes. Nothing is wrong in any of these samples — the only thing moving is luck.

Each draw gives one test statistic, and the chart collects them. Our own audit gave 2.40, the orange line.

Draw a few one at a time, then draw a lot. Where does 2.40 sit?

Draw 1 sample | Draw 25 | Draw 500 | Start over | ☐ show the normal curve

samples drawn: 0 of those, this far from 0 or further: 0 (–)


our audit: 2.40
press a button to draw a sample
-4
-3
-2
-1
0
1
2
3
4
test statistic from one sample of 36 deliveries

What you just built

That chart is the distribution of the test statistic when H_0 is true. It is our benchmark: it shows which values are ordinary when the courier is doing nothing wrong.

Two things are now visible:

- samples from an on-time courier pile up around 0, and almost never reach 2.4;
- our audit gave 2.4, out in the thin part.

But you only ever get one sample

We just drew 500 samples. In real life you draw one.

You time 36 deliveries, once. You cannot repeat the audit 500 times to find out which statistics are ordinary and which are strange.

You do not need to. We drew samples here only so you could see the picture. Its shape can be worked out theoretically — on paper, before collecting any data — as soon as you assume H_0 is true.

So the benchmark is something you know in advance, not something you measure. Which distribution it is depends on how large your sample is, and we come back to that shortly.

So how do we decide?

Take the test statistic and compare it with a cutoff.

1. Choose a cutoff far out in that picture — a value we would almost never see if the courier were on time.
2. Compare our 2.40 with it.
3. Past the cutoff → reject H_0 and claim the discount. Not past it → we cannot claim.

Why far out? If the courier really did average 100 minutes, a sample that extreme would almost never happen.

So if we actually observe one, that is evidence against 100 being the true mean.

Either we were very unlucky, or the claim of 100 is wrong. Past the cutoff we take the second explanation.

For example, put the cutoff at 1.645. In the picture you just drew, only about 5 samples in 100 reach that far when the courier is on time.

Our audit gave 2.40, which is past 1.645 → reject H_0, claim the discount.

Had the audit come back at 1.2, we would not have claimed: 1.2 is the kind of number an on-time courier produces all the time.

Two names for the rest of the week:

- the cutoff and everything beyond it → the rejection region
- the share of the benchmark beyond our 2.40 → the p-value

Everything that remains is about where to put the cutoff — and what it costs you when you put it in the wrong place.

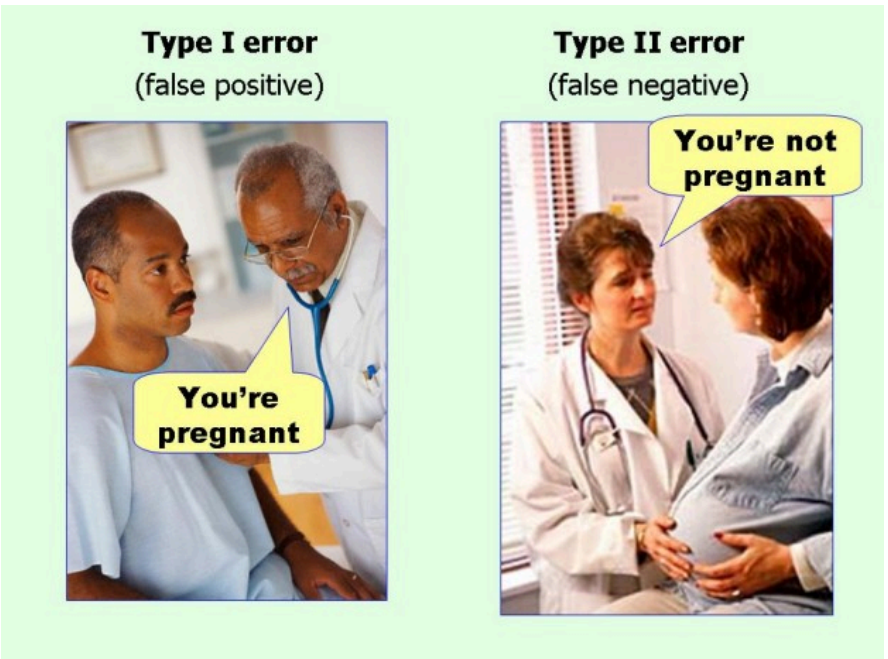
3 · The two ways to be wrong

Every cut-off trades one error against the other. Power is what you buy with that trade.

Decision errors

	Do not reject H_0	Reject H_0 in favor of H_A
H_0 true	Okay	Type 1 Error
H_A true	Type 2 Error	Okay

- H_0: You are not pregnant
- H_1: You are pregnant



Decision errors: free shipping

Do people spend more if they get free shipping? (H_0: they spend the same)

- **Type 1 error**: reject null when H_0 is true
  - Conclude free-shipping customers spend more, while in reality they spend the same
- **Type 2 error**: don't reject null when H_A is true
  - Conclude they spend the same, while free-shipping customers actually spend more
- Trial of a murderer: H_0 innocent, H_1 guilty. Describe error types 1 and 2.
- Type 1: Convict an innocent
- Type 2: Do not convict a murderer
- Probability of a type 1 error is α — the **size** or **significance level**.
  - We don't want to incorrectly reject the null more than $\alpha \cdot 100\%$ of the time.
- Probability of a type 2 error is β — to calculate it, you need the true μ.

Probability of type 1 error

In words: how often do we claim the discount from a courier who is actually on time?

$$\alpha = P(\text{reject } H_0 \mid H_0 \text{ is true})$$

- In the courier example, suppose the null is true and the courier really does average $\mu = 100$ minutes.
- You take a sample of 36.
- Say you were unlucky: by chance your 36 deliveries hit bad traffic and $\bar{x} = 106$, so $z = \frac{106-100}{20/\sqrt{36}} = 1.8$.
- You are using the cutoff of 1.645, and $1.8 > 1.645$, so you claim the discount — but the courier was meeting the contract all along. An error.
- Q: What is the probability the standardized mean of these 36 is in the rejection region (larger than 1.645)?
- What's the probability of making such an error?

$$\alpha = P_{H_0}(Z > 1.645)$$

$$= 1 - \Phi(1.645) = 0.05$$

The whole calculation lives on the test-statistic scale. Under H_0 the statistic is standard normal, so the area above the cutoff is α — nothing else to compute.

Move the cutoff and watch α

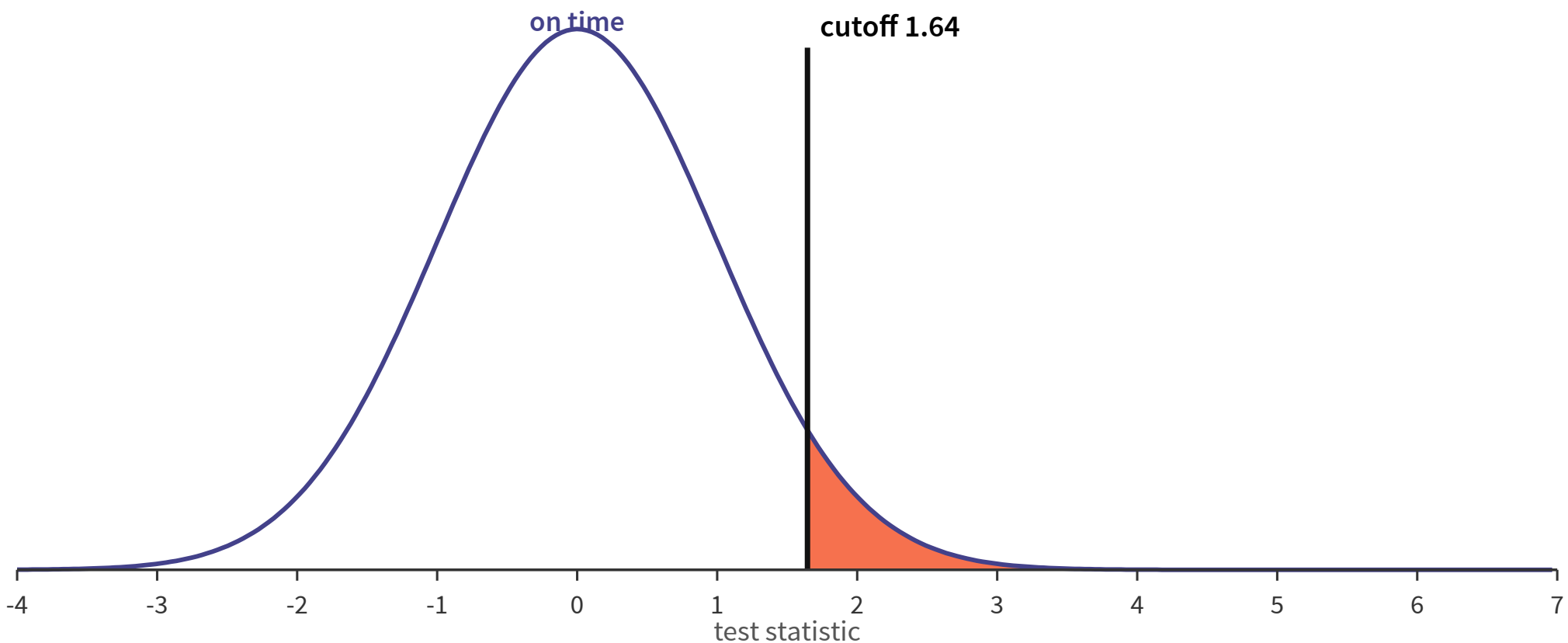
The curve is the benchmark from before: what the test statistic does when the courier is on time. The shaded tail is the rejection region.

Move the significance level and watch two things: where the cutoff sits, and how wide the shaded tail is.

significance level α **α = 0.0500** ↔ cutoff **cutoff z = 1.645**

☑ courier is on time ☐ courier is really late true mean 110 min sample size n 36

reject when the sample mean exceeds 105.5 min



Probability of type 2 error

In words: how often do we fail to claim from a courier who really is late?

$$\beta = P(\text{fail to reject } H_0 \mid H_A \text{ is true})$$

- β is just an area under a curve — the part of the distribution that lands below the cutoff, where we fail to claim.
- We know how to find areas under a curve: that is what `pnorm()` does.
- The one new thing: this curve is not centred at 0. The courier really is slow, so the whole distribution sits further to the right.
- So we do the only thing we need to do — measure how far right it sits, and shift by that much.
- And to measure that, we first have to say how late. Say the courier really averages $\mu_1 = 110$ minutes.

Trade-off between type 1 and 2 errors

$$\alpha = 1 - \Phi(z_{1-\alpha}) \qquad \beta = \Phi(z_{1-\alpha} - d)$$

Both formulas contain the same critical value $z_{1-\alpha}$. Moving it moves both numbers — in which directions?

Power of a test

- Power is the probability of rejecting if the alternative is true.
- Power differs for every possible value of the alternative.
- Power for a one-sided test:

$$Power = P_{H_A}(Reject) = 1 - \underbrace{P_{H_A}(\text{Not Reject})}_{\text{type 2 error}} = 1 - \beta$$

$$\text{Power} = 1 - \Phi(z_{1-\alpha} - d)$$

What drives power

Power increases if: n increases; the alternative is further from the null; σ decreases; α increases.

The first two are what you just measured; the last one is the trade-off from Exercise 9, seen from the other side.

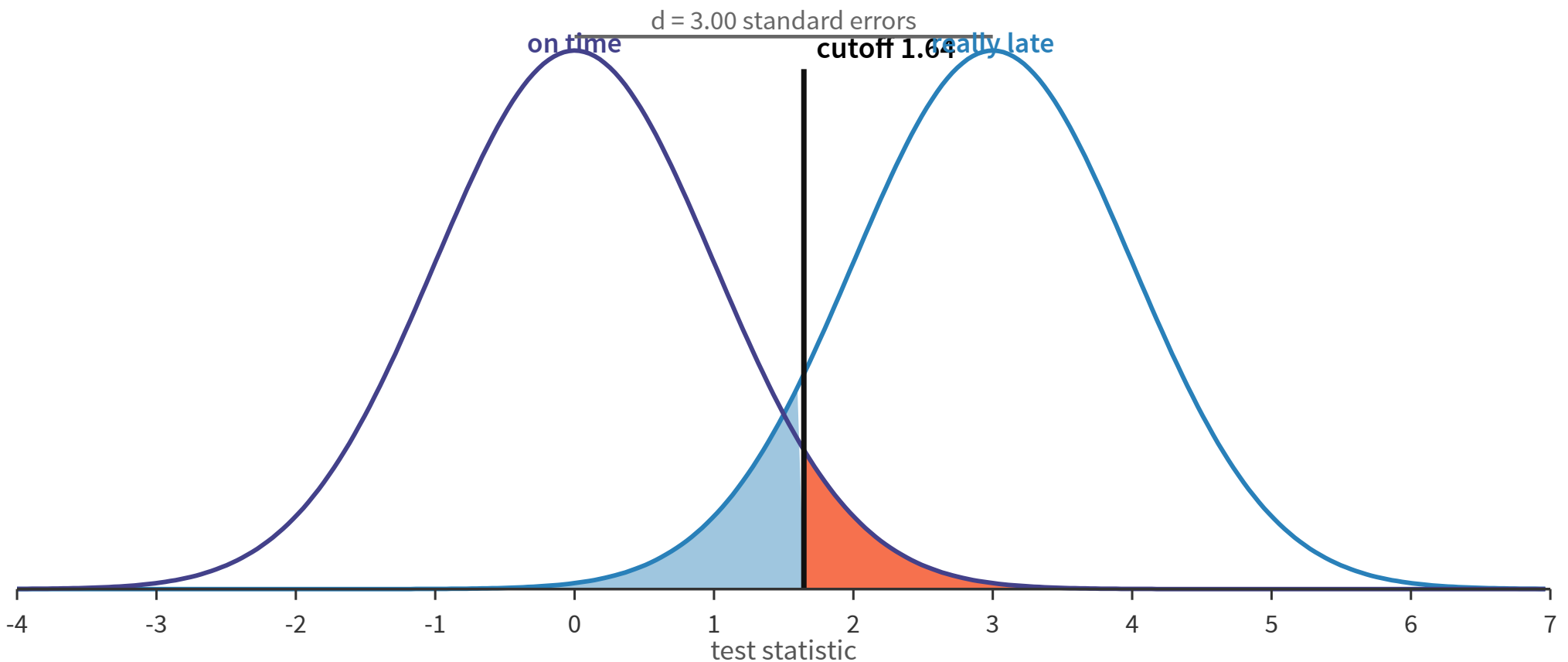
Move the cutoff and watch α — and power

Both curves are on. Move one control at a time and watch power: the true mean, the sample size n, the standard deviation σ, and the significance level α.

significance level α **α = 0.0500** ↔ cutoff **cutoff z = 1.645**

courier is on time **courier is really late** true mean 110 min sample size n 36 σ 20 min

reject when the sample mean exceeds 105.5 min **β = 0.088** **power = 0.912**



4 · The test for one mean

The procedure, run end to end, on the delivery claim.

Single Sample Test for Mean

General idea: test if the parameter equals/exceeds/falls below some value.

- Ex: $H_0 : \mu = 3$ vs $H_A : \mu \neq 3$
- **Test statistic**: normalized sample mean

$$\text{test statistic} = \frac{\bar{X} - \mu_0}{SE}$$

- SE depends on the estimate of the standard deviation:
  - Standard deviation known: $SE = \frac{\sigma}{\sqrt{n}}$
  - Standard deviation not known: $SE = \frac{s}{\sqrt{n}}$
- **Rejection region**: depends on the distribution
  - Large sample or known variance: $Z_{test} \sim N(0, 1)$ — critical values from standard normal
  - Small sample, normal data: $T_{test} \sim t(n-1)$ — critical values from Student's t, $n - 1$ df

Single Sample - Mean - Two Sided Test

Hypothesis: $H_0 : \mu = \mu_0$ and $H_A : \mu \neq \mu_0$

Rejection Region: for a test of significance level α, reject

- If large sample or known variance from normal

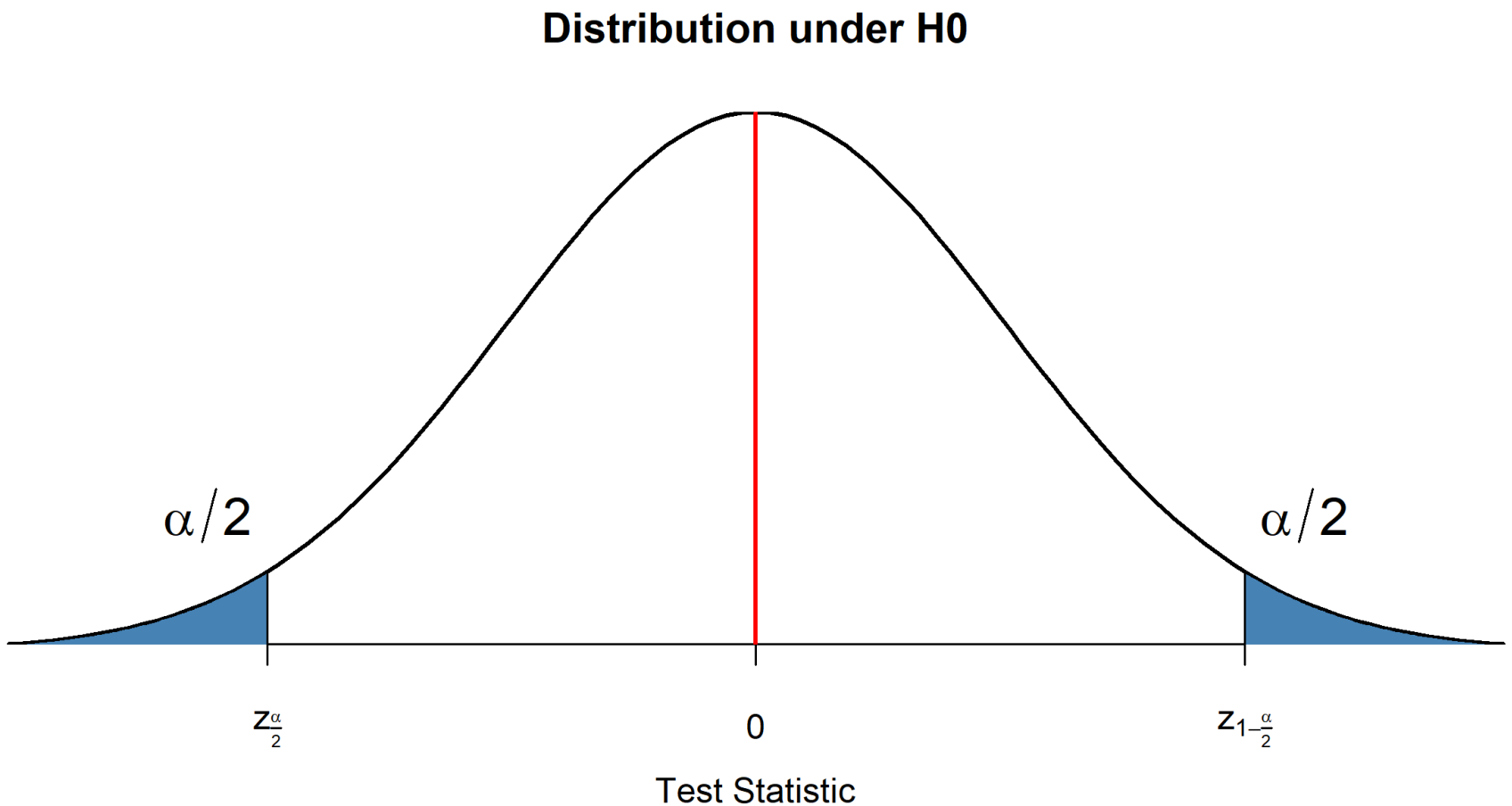
$$\frac{\bar{X} - \mu_0}{SE} < z_{\frac{\alpha}{2}} \qquad or \qquad \frac{\bar{X} - \mu_0}{SE} > z_{1-\frac{\alpha}{2}}$$

- If small sample from normal

$$\frac{\bar{X} - \mu_0}{SE} < t_{(n-1),\frac{\alpha}{2}} \qquad or \qquad \frac{\bar{X} - \mu_0}{SE} > t_{(n-1),1-\frac{\alpha}{2}}$$

Where $z_{1-\frac{\alpha}{2}}$ and $t_{(n-1),1-\frac{\alpha}{2}}$ are $(1 - \frac{\alpha}{2})$ quantiles of the standard normal and Student's t with $n - 1$ df.

Single Sample - Mean - Two Sided Test: the picture



Single Sample - Mean - One sided - Case 1

Hypothesis: $H_0 : \mu = \mu_0$ and $H_A : \mu < \mu_0$

- This includes any null like $\mu \geq \mu_0$. Why?
- If you rejected μ_0 at α, you would reject for sure anything larger than μ_0. See it move.

Rejection Region: for a test of significance level α, reject

- If large sample or known variance from normal

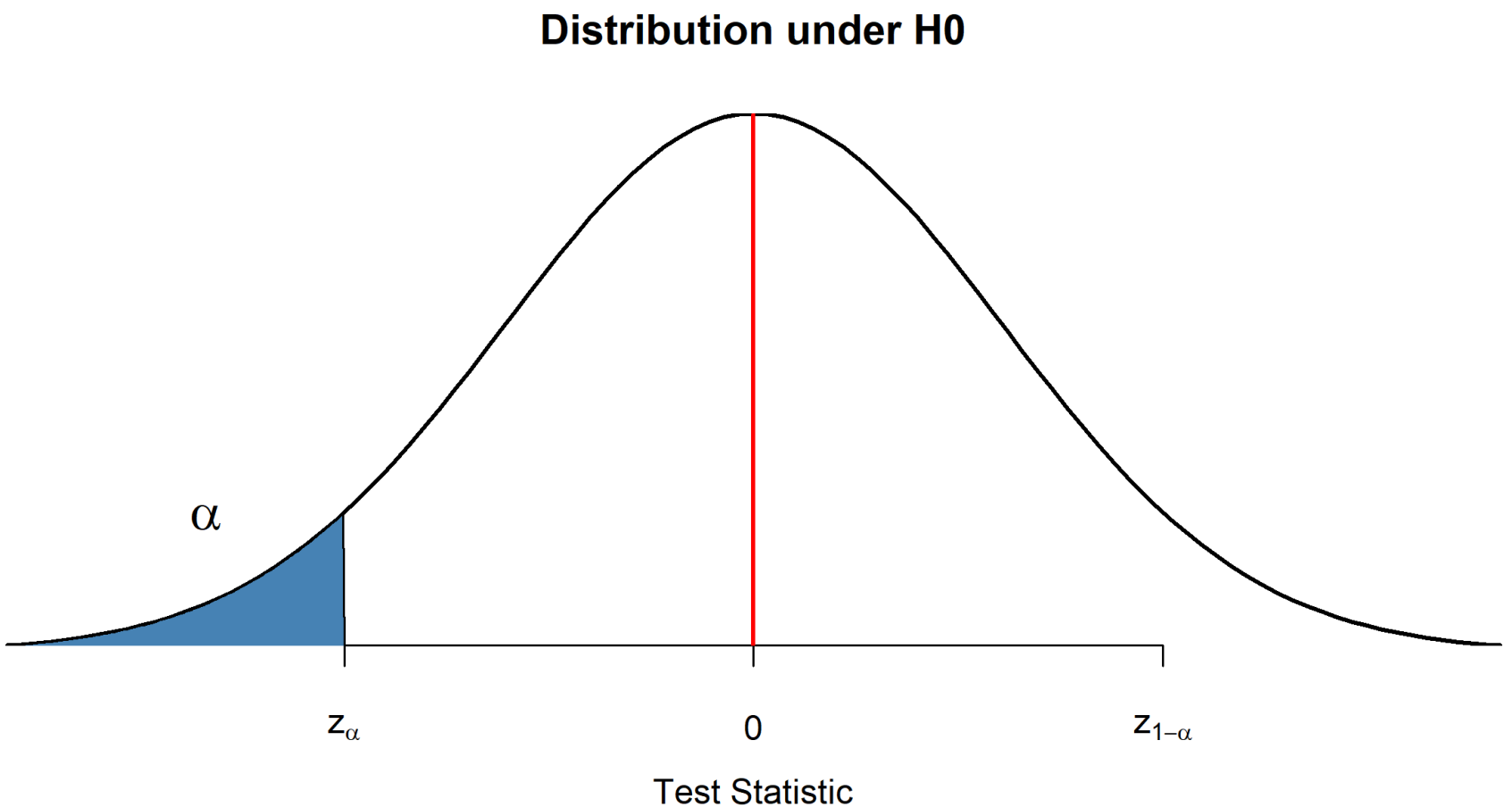
$$\frac{\bar{X} - \mu_0}{SE} < z_\alpha$$

- If small sample from normal

$$\frac{\bar{X} - \mu_0}{SE} < t_{(n-1),\alpha}$$

Where z_α and $t_{(n-1),\alpha}$ are α quantiles of the standard normal and Student's t with $n - 1$ df.

Single Sample - Mean - One sided - Case 1: the picture



Case 2 — the mirror image

Hypothesis: $H_0 : \mu = \mu_0$ and $H_A : \mu > \mu_0$ — this covers any null like $\mu \leq \mu_0$.

- Everything flips to the right tail. Reject when

$$\frac{\bar{X} - \mu_0}{SE} > z_{1-\alpha} \qquad (\text{or } t_{(n-1),1-\alpha} \text{ for a small sample})$$

- This is the courier test: $H_A : \mu > 100$. The launch decision next is Case 2 too.

Full slides for Case 2

The launch decision

A fuel-card company is deciding whether to launch a national loyalty programme. Finance say it only breaks even if Mexican households spend more than 200 pesos a week on gasoline, on average.

You have the Households & Environment Survey — real data, 14,505 households, in `exam_data$gasoline_spending_weekly`.

Should they launch?

Exercise 19 — The whole test, start to finish

Work every step yourself.

1. Write H_0 and H_A. The company acts only on evidence of "more than 200" — so which side is the alternative?
2. The sample mean, the standard deviation, and the standard error.
3. The test statistic.
4. The 5% cutoff, and your decision.
5. The p-value. Would you still launch at $\alpha = 0.01$?
6. Finally, compute `median(gas)`. Then write one sentence for the board.

Exercise | Start Over | Show Hint | Run Code

```
1  gas <- exam_data$gasoline_spending_weekly
2  n   <- 14505
```

5 · P-values

The same decision, reported as a number instead of a yes or no.

P-values

Definition: the p-value is the probability, under the null hypothesis, of obtaining a test statistic at least as extreme as the one computed from the sample.

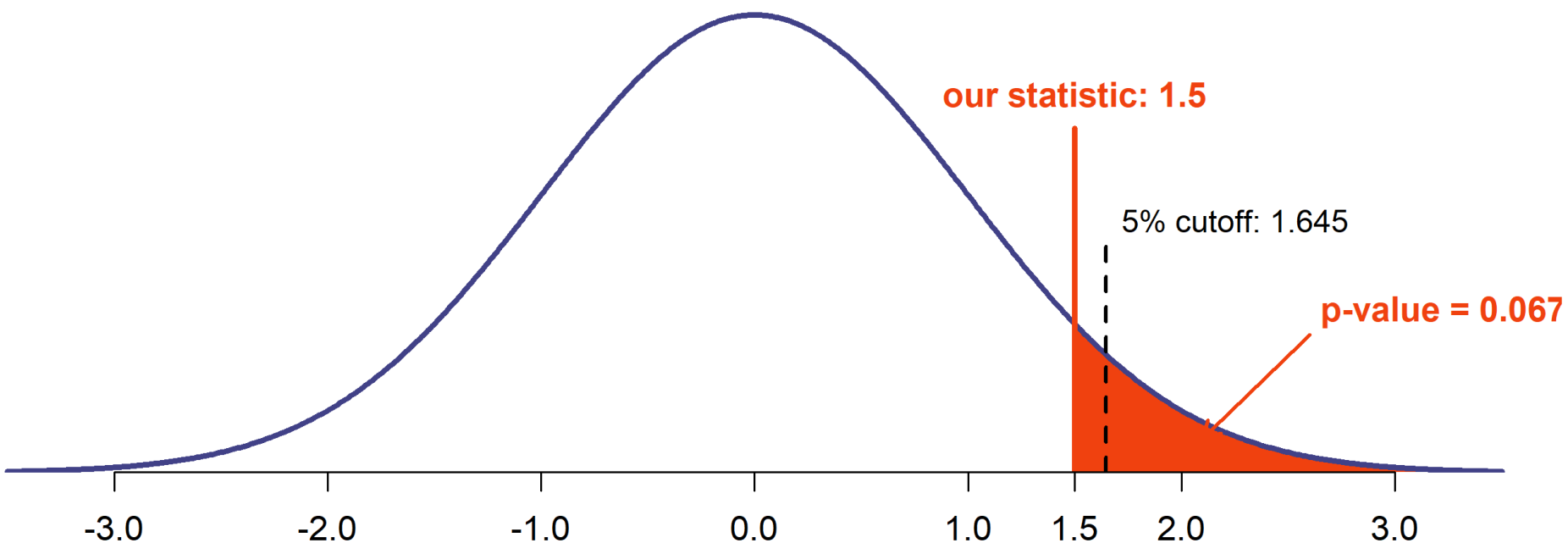
Depends on: your hypothesis, and the calculated test statistic.

- Suppose under H_0 the test statistic is $Z_{test} \sim N(0, 1)$.
- One-sided tests:
  - $H_A : \mu > \mu_0$: $p\text{-value} = P(Z_{Test} \geq \text{computed statistic})$
  - $H_A : \mu < \mu_0$: $p\text{-value} = P(Z_{Test} \leq \text{computed statistic})$
- Two-sided test:
  - $H_A : \mu \neq \mu_0$: $p\text{-value} = 2P(Z_{Test} \geq |\text{computed statistic}|)$

P-values: Example

- Test $H_0 : \mu = 5$ vs $H_A : \mu > 5$.
- In a sample of 36 we have $\bar{x} = 5.5$ and $s = 2$.
- The test statistic is distributed normally:

$$p\text{-value} = P(Z > \frac{5.5 - 5}{2/6}) = P(Z > 1.5) = 0.067$$



P-value in two-sided test

- In a two-sided test, the p-value accounts for both tails.
- We double the one-tail probability: $p\text{-value} = 2P(Z \geq |z_{test}|)$.

Interactive visualization: https://rpsychologist.com/pvalue/

Exercise 16 — The p-value from the same data

a. Compute the two-sided p-value for the courier test two ways from your hand-built statistic: once with `pnorm()`, once with `pt()`. Which of the two is the right reference here, and how much does the choice matter?

Exercise | Start Over | Show Hint | Run Code

```
1 mins <- deliveries$minutes
2 n    <- 36
```

b. The contract is one-sided — you only claim a discount if the courier is slower. Compute the one-sided p-values from that same statistic, for $H_A : \mu > 100$ and for $H_A : \mu < 100$, and state how the three p-values relate to each other.

Exercise | Start Over | Show Hint | Run Code

```
1 mins <- deliveries$minutes
2 n    <- 36
```

What a p-value actually counts

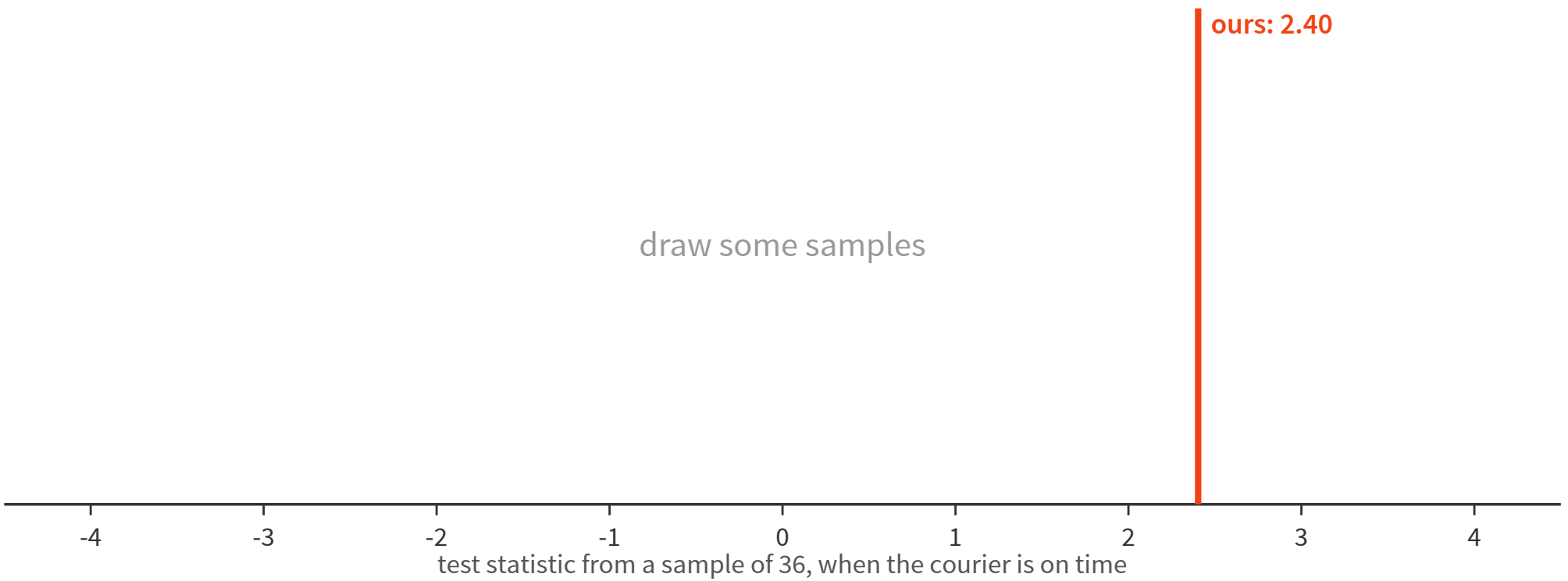
The cutoff asked a yes/no question. The p-value asks a sharper one:

> if the courier is on time, how often would luck alone hand us a statistic at least as extreme as the one we got?

Draw samples from an on-time courier. Every sample that lands at or beyond our 2.40 gets counted. The p-value is the share that do.

Draw 1 | Draw 50 | Draw 1000 | Start over | ☐ two-sided (count both tails)

samples drawn: 0 **at least as extreme as 2.40: 0** **share = –**



Two things to notice as the count grows:

- the running share settles near 0.011, the one-sided p-value — it is a *proportion of samples*, nothing more exotic;
- the p-value does not depend on α. It is a property of the data. α only enters when you compare the two and decide.

Switch to two-sided and the share roughly doubles: now a sample counts if it is that far from 0 in either direction, and the on-time curve is symmetric.

P-values and Test Significance

- In our example we calculated the p-value of 0.067.
- Would a test at $\alpha = 0.05$ reject the null?
- Any test with significance level $\alpha > p\text{-value}$ would reject the null.

Another way to think about p-values:

- The p-value is the smallest significance level α at which H_0 can be rejected.

(3) The average grade among fourth graders on a math aptitude test in a certain state is 62.5 out of 100 points. (Assume 62.5 is the population average.) An educational company has developed an app for elementary-school math and would like to test whether the app improves average performance on the aptitude test among fourth graders. Let μ_X denote the population mean of test scores among students who use the app.

(a) To provide statistical evidence that the app *increases* the true average of test scores, what are the appropriate one-sided null hypothesis and alternative hypothesis?

(b) The company provides the app to 15 randomly selected students. The sample mean and sample standard deviation of test scores are 68.3 and 13.5, respectively. Under the assumption that the test scores are i.i.d. and normally distributed, would the t-test of the null hypothesis in (a) be rejected at the 5% level? at the 10% level?

(c) What is the p-value associated with the t-test in (b)?

Exercise 17 — Decision at α = 0.05

Your two-sided test on the delivery times returned the p-value you computed in Exercise 16. Testing at $\alpha = 0.05$, what do you conclude?

- Reject H_0 — at $\alpha = 0.05$ there is evidence the mean delivery time differs from 100
- Fail to reject H_0 — p is above 0.05
- Accept H_0 — the mean equals 100 exactly
- Nothing can be decided without knowing the sample size

Reject H_0 — at $\alpha = 0.05$ there is evidence the mean delivery time differs from 100.

Rule: reject H_0 when p-value < α. Here $0.022 < 0.05$, so we reject — though at $\alpha = 0.01$ the same data does not reject (Exercise 14). We never "accept" H_0; we only reject or fail to reject.

6 · One procedure, four tests

The five steps never change. Only the statistic and its reference distribution do.

The same five steps every time

1. State H_0 and H_1.
2. Compute the statistic: (estimate − claim) / standard error.
3. Find its distribution when H_0 is true.
4. Compare against the cut-off, or compute the p-value.
5. Decide, and say what it means for the business.

Steps 1, 4 and 5 are identical in every test you will meet. Step 2 changes the estimate and the standard error. Step 3 changes the degrees of freedom.

What changes from test to test

Test	H_0	Statistic	Reference
One mean	$\mu = \mu_0$	$\dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$	t_{n-1}
Two means	$\mu_X - \mu_Y = 0$	$\dfrac{\bar{x} - \bar{y}}{\sqrt{s_X^2/n + s_Y^2/m}}$	t_v, Welch v
Paired	$\mu_d = 0$	$\dfrac{\bar{d}}{s_d/\sqrt{n}}$	t_{n-1}
Correlation	$\rho = 0$	$\dfrac{\hat{\rho}}{\sqrt{(1-\hat{\rho}^2)/(n-2)}}$	t_{n-2}

- Every statistic has the same shape: how far from the claim, divided by how much it wobbles.
- Only the middle two columns change. Steps 1, 4 and 5 are the same every time.
- Use the standard normal instead of t once the sample is large.

Which test does the question call for?

The question says	The test
One group, one claimed value	One mean
Two separate groups of units	Two means
The same units measured twice	Paired
Two columns, do they move together	Correlation

The trap is the third row. Two columns of numbers are paired only when row i is the same unit in both. If the rows are different people, it is a two-sample problem.

Choosing the Appropriate Test

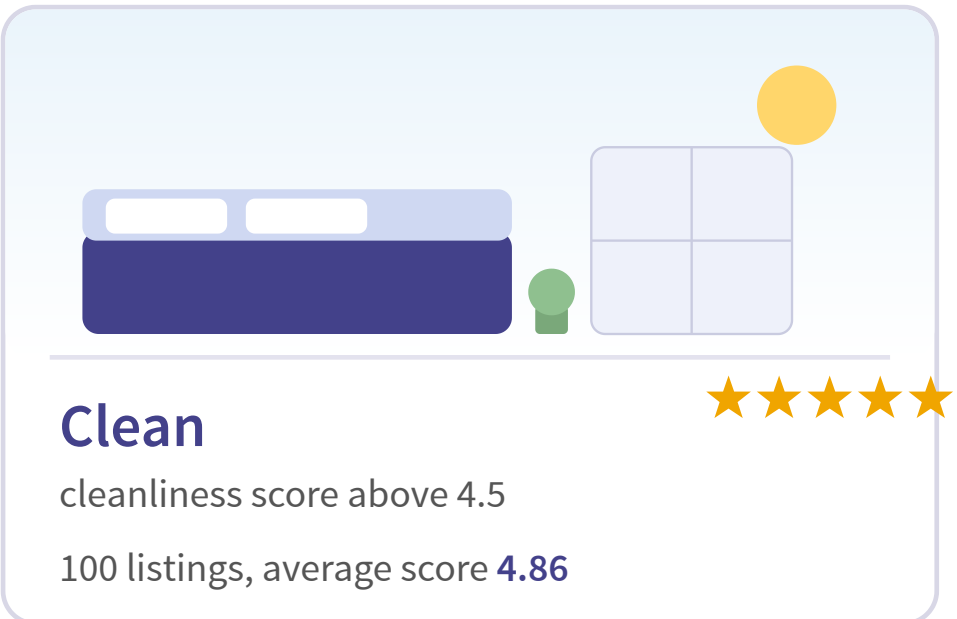
The choice of test statistic and rejection region depends on:

- **Type of data we are testing?**
  - Single sample: $H_0 : \mu = \mu_0$ vs $H_A : \mu \neq \mu_0$
  - Two independent samples: $H_0 : \mu_{s1} = \mu_{s2}$ vs $H_A : \mu_{s1} \neq \mu_{s2}$
  - Paired data: $H_0 : \mu_{d1} = \mu_{d2}$ vs $H_A : \mu_{d1} \neq \mu_{d2}$
- **Can we estimate the standard deviation well?**
  - Yes: large sample, or known standard deviation
  - No: small sample but normal distribution

7 · Two means

Two groups of listings, and whether the gap between them is bigger than luck.

Can we charge more for a clean apartment?

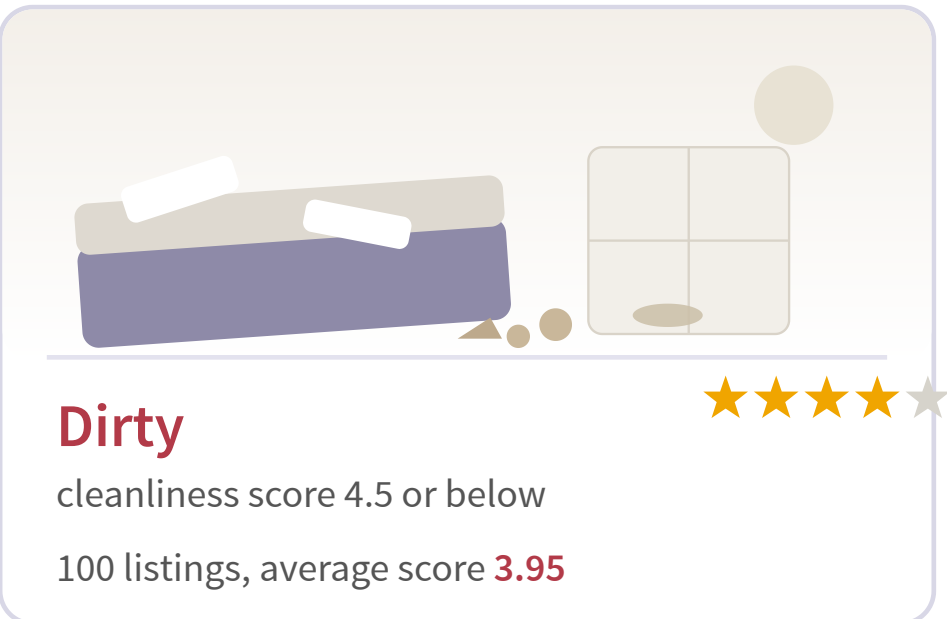


★★★★★

Clean

cleanliness score above 4.5

100 listings, average score **4.86**



★★★★★

Dirty

cleanliness score 4.5 or below

100 listings, average score **3.95**

You run Airbnbs in Mexico City: 200 listings, split into Clean (cleanliness score above 4.5) and Dirty.

- clean listings average 1,245 pesos a night
- dirty listings average 869 pesos a night

A gap of 377 pesos — in these 200 listings.

Is the true mean price of clean apartments higher than the true mean price of dirty ones?

Why the gap alone settles nothing

You are not comparing two numbers. You are comparing two samples.

Take any 200 listings, split them into two halves at random, and the two averages will still come out different. Never exactly 0.

So the question is not “is there a gap?” — there always is. It is:

is 377 pesos bigger than the gap luck alone would have produced?

And now both sides carry noise, so what we compare against has to account for both.

Two samples and the difference in means

Two iid samples, independent of each other, from two populations $X_1, \ldots, X_n$ and $Y_1, \ldots, Y_m$.

General idea: test the difference between population means.

- $H_0 : \mu_X - \mu_Y = 8{,}000$ vs $H_A : \mu_X - \mu_Y \neq 8{,}000$
  - a health insurer charges smokers 8,000 pesos a year more than non-smokers, because its model says smokers cost that much more in claims. X = yearly claims of a smoker, Y = yearly claims of a non-smoker. Is the difference in mean claims really 8,000? If it is smaller, smokers are overcharged and move to a competitor. If it is larger, the insurer loses money on every smoker it covers.

- $H_0 : \mu_X - \mu_Y = 0$ vs $H_A : \mu_X - \mu_Y > 0$
  - our apartments, with X = the price of a clean listing and Y = a dirty one: "cleanliness does not change the price" against "clean listings are more expensive" — this is our question

- Test for no difference: $H_0 : \mu_X - \mu_Y = 0$ vs $H_A : \mu_X - \mu_Y \neq 0$
  - same null, but now a gap in either direction counts as evidence

- Test statistic: normalized difference in sample means

$$\text{test stat} = \frac{\bar{X} - \bar{Y} - \overbrace{(\mu_{X,0} - \mu_{Y,0})}^{\Delta_0}}{SE} = \frac{\bar{X} - \bar{Y} - \Delta_0}{SE}$$

How much does the gap bounce around?

We have the gap $\bar{X} - \bar{Y}$. To judge it we need to know how far it wanders from sample to sample.

Each mean carries its own noise:

$$Var(\bar{X}) = \frac{\sigma_X^2}{n} \qquad\qquad Var(\bar{Y}) = \frac{\sigma_Y^2}{m}$$

The two samples are independent, so the variances add — even though we subtract the means:

$$Var(\bar{X} - \bar{Y}) = Var(\bar{X}) + Var(\bar{Y}) = \frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}$$

The standard error is the square root of that:

$$SE = \sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}$$

We almost never know σ_X and σ_Y, so we use what the samples give us:

$$\widehat{SE} = \sqrt{\frac{s_X^2}{n} + \frac{s_Y^2}{m}}$$

Careful: variances add, standard errors do not. Never write $SE_X + SE_Y$, and never subtract them just because the means were subtracted.

Which gaps count as too big?

Put the pieces together and the statistic is

$$t = \frac{\bar{X} - \bar{Y} - \Delta_0}{\widehat{SE}}$$

Suppose it comes out at 3.2. Is that too big?

We cannot say yet. A number on its own decides nothing.

We first need to know how this statistic behaves when H_0 is true — which values are ordinary, and which almost never happen.

What the statistic does when H_0 is true

The CLT gives each sample mean a normal shape:

$$\bar{X} \sim N\left(\mu_X,\ \frac{\sigma_X^2}{n}\right) \qquad and \qquad \bar{Y} \sim N\left(\mu_Y,\ \frac{\sigma_Y^2}{m}\right)$$

Subtract two independent normals and you get a normal again — centred on the true gap, with the variance we just built:

$$\bar{X} - \bar{Y} \sim N\left(\mu_X - \mu_Y,\ \frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}\right)$$

If H_0 is true that centre is Δ_0. Subtract it, divide by the standard error:

$$\frac{\bar{X} - \bar{Y} - \Delta_0}{\widehat{SE}} \sim N(0,1)$$

What if the samples are small?

We swapped σ_X, σ_Y for s_X, s_Y. With plenty of data that costs nothing.

With few observations s is itself noisy. Sometimes it comes out too small, which makes the statistic too big — so extreme values happen more often than a normal would predict.

Fatter tails. So we compare against Student's t instead, with v degrees of freedom.

How many degrees of freedom?

Both groups here hold 100 listings, so $n = m = 100$. Two ways to get v:

1. Proper, complex (and annoying):

$$v = \frac{\left(\frac{s_X^2}{n} + \frac{s_Y^2}{m}\right)^2}{\frac{(s_X^2/n)^2}{n-1} + \frac{(s_Y^2/m)^2}{m-1}}$$

Round down to the nearest integer. For the apartments this needs s_X and s_Y as well — the two price standard deviations — so we cannot fill it in until we have looked at the data.

2. Simple, approximate shortcut:

$$v = min(n-1, m-1)$$

For the apartments that is $\min(99,\ 99) = 99$ — the sample sizes alone are enough. In Exercise 8 you will see how far it sits from the proper number.

Meet the data

`Sample_list` holds the 200 Mexico City listings from the opening question. Three columns matter:

- `price` — the nightly price in pesos
- `review_scores_cleanliness` — the guest cleanliness rating, 1 to 5
- `clean` — `TRUE` when that rating is above 4.5, `FALSE` otherwise

Run this to see the two groups. Press Run — R runs in your browser.

Exercise | Start Over | Show Solution | Run Code

```
1 Sample_list[1:6, ]
2
3 table(Sample_list$clean)
```

Exercise 2 — The whole two-sample test

Test whether clean listings are priced differently from dirty ones: $H_0 : \mu_{clean} = \mu_{dirty}$ against $H_A : \mu_{clean} \neq \mu_{dirty}$. Work every step.

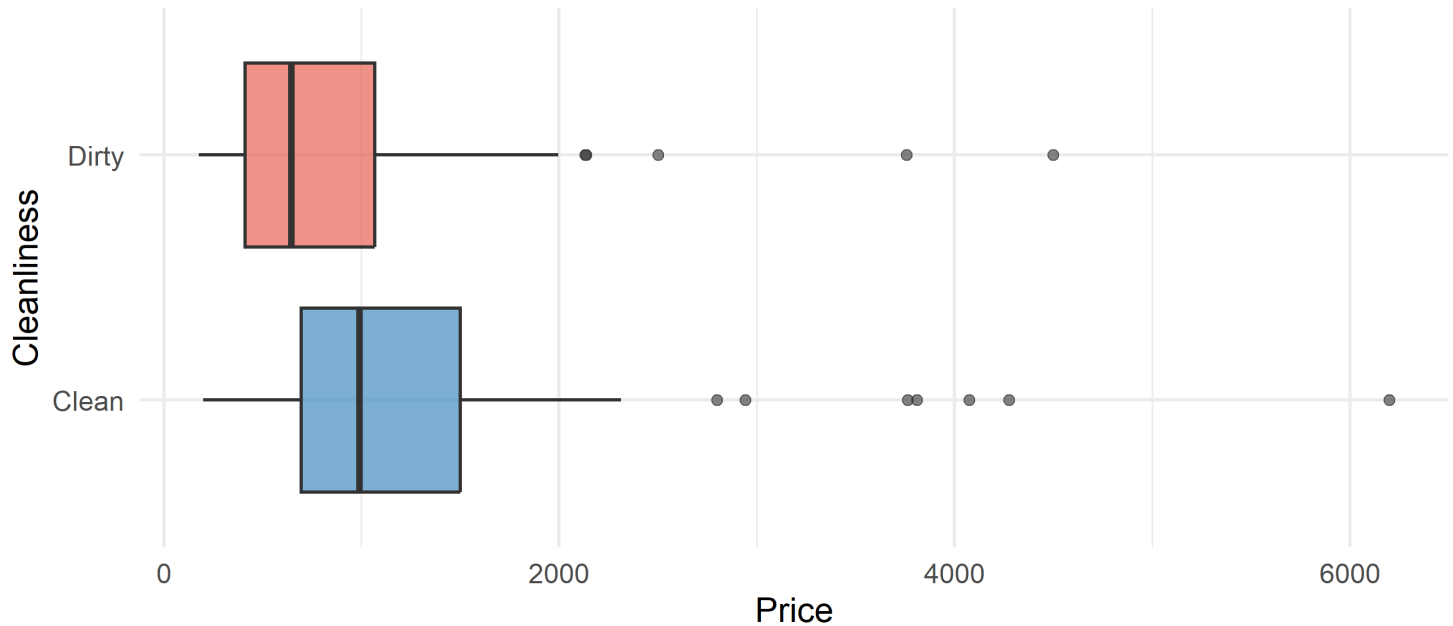
1. The two group means, and the gap between them.
2. The standard error of that gap.
3. The test statistic.
4. The degrees of freedom, using the shortcut $v = \min(n-1, m-1)$.
5. The two-sided cut-off at $\alpha = 0.05$, and your decision.
6. The two-sided p-value.

Exercise | Start Over | Show Hint | Run Code

```
clean <- Sample_list$price[Sample_list$clean == TRUE]
dirty <- Sample_list$price[Sample_list$clean == FALSE]
n <- 100
m <- 100
```

The two groups, side by side

	n	Mean	SD
Clean	100	1245.4	961.9
Dirty	100	868.7	693.3



- The clean group's standard deviation (962) is *larger* than the dirty one's (693).
- The boxplot shows why: a long right tail of very expensive clean listings.
- Two groups with different standard deviations is the normal case — which is exactly why the standard error keeps them separate instead of pooling them.

Pooled or Welch?

- Pooled ($v = n + m - 2$) assumes $\sigma_X = \sigma_Y$ and averages the two sample variances into one. Cheap, and wrong when the two standard deviations differ.
- Welch makes no such assumption, and pays for it with a fractional v. It is the safe default whenever the two standard deviations might differ.
- The exam shortcut $v = \min(n - 1, m - 1)$ is deliberately conservative: fewer df → fatter tails → you reject slightly less often than you should. Safe, never anti-conservative.

Exercise: McDonald's vs Burger King

We suspect Burger King has larger caloric content for kids' meals.

Sample	n	Sample Mean	Sample Standard Deviation
McDonald	9	475	230
Burger King	16	520	180

- State the null and alternative hypothesis
- Calculate the value of the test statistic
- Determine the appropriate number of degrees of freedom
- Calculate the p-value; at which α do we reject?

🖊️ Exercise 9 — Small samples: does the df choice matter?

Same job as Exercise 8, but now $n = 9$ and $m = 16$ — small enough that the t distribution really bites.

1. Write H_0 and H_A for the claim "Burger King meals have more calories".
2. Compute the SE and the test statistic.
3. Compute both degrees of freedom: the Welch formula and $\min(n-1, m-1)$.
4. Compute the one-sided p-value under each. How far apart are they?
5. At $\alpha = 0.05$, does the choice of df change your decision?

Exercise | Start Over | Show Hint | ▷ Run Code

```r
n_m <- 9;  mean_m <- 475; sd_m <- 230   # McDonald's
n_b <- 16; mean_b <- 520; sd_b <- 180   # Burger King

A <- sd_m^2/n_m
B <- sd_b^2/n_b

# 2. the standard error and the test statistic

# 3. the two degrees of freedom

# 4. the one-sided p-value under each
```

8 · Paired data

The same units measured twice. Pairing removes the differences between units.

Paired data

- Sometimes we have only one set of individuals or objects, but two (or more) observations per each.
- Example: employee productivity — two observations per employee: productivity working from home and working from the office.
- Example: sales at Walmart and Soriana in each neighborhood — two observations per neighborhood.

Paired data — setup

We have a set of independent pairs of data $(X_1, Y_1), (X_2, Y_2), \ldots, (X_n, Y_n)$.

Observations within a pair are not necessarily independent!

- Some productive workers are productive both at home and in the office; some unproductive are unproductive in both.

- We observe the difference for each individual: $X_i - Y_i = d_i$.
- We look at the population value of that difference: $\mu_X - \mu_Y = \Delta$.
  - Check if the means are the same: $\Delta = 0$ (no difference in productivity)
  - Or mean X is bigger than mean Y: $\Delta > 0$

Paired data — test statistic

Null hypothesis: $H_0 : \Delta = \Delta_0$

Alternative: $H_A : \Delta \neq \Delta_0$ (two-sided), or $H_A : \Delta > \Delta_0$, or $H_A : \Delta < \Delta_0$ (one-sided)

Test Statistic is:

$$\text{test statistic} = \frac{\overbrace{\bar{X} - \bar{Y}}^{\bar{d}} - \Delta_0}{SE} = \frac{\bar{d} - \Delta_0}{SE}$$

- Note on the standard error — it's not the same formula as for two independent samples: $Var(X_i - Y_i) \neq Var(X_i) + Var(Y_i)$
- We just calculate it from the differences:
  - If we know the variance, $SE = \frac{\sqrt{var(d_i)}}{\sqrt{n}} = \frac{\sigma_d}{\sqrt{n}}$
  - If we don't: $SE = \frac{s_d}{\sqrt{n}}$

Paired or unpaired data?

1. We want to know if Intel's stock and Southwest Airlines' stock have similar rates of return. We take a random sample of 50 days and record both stocks on those same days.

- Paired

2. We randomly sample 50 items at Target and note each price. Then we visit Walmart and collect the price for each of those same 50 items.

- Paired

3. A school board wants to know if average SAT scores differ between two high schools. They take a simple random sample of 100 students from each school.

- Not paired

Same 40 employees, measured twice

`productivity` holds 40 employees, each with an `office` score and a `home` score: the same person appears in both columns.

Some people are simply more productive than others — at home *and* at the office. That person-level variation is in both columns, and it is not what you are trying to measure.

The question is whether working from home changes output. Two ways to ask it:

- treat the 80 numbers as two independent samples of 40, or
- work with the 40 differences `home - office`.

Same 80 numbers. Should the two routes agree?

Exercise 18 — Ignore the pairing, then use it

1. Look at the first few rows of `productivity`, and the correlation between the two columns.
2. Build the unpaired standard error from `sd(productivity$home)` and `sd(productivity$office)`, and the statistic that goes with it.
3. Now build `d <- productivity$home - productivity$office`, its standard error `sd(d)/sqrt(40)`, and the paired statistic.
4. Compare the two standard errors. Which is smaller, and by how much?

Exercise | Start Over | Show Hint | Run Code

```
1 productivity[1:6, ]
```

Why pairing wins

- $Var(X_i - Y_i) = Var(X_i) + Var(Y_i) - 2\,Cov(X_i, Y_i)$.
- Positively correlated pairs → that last term is large and positive → the variance of the difference collapses.
- The independent-samples formula silently sets $Cov = 0$ and inflates the SE.
- Cost: the paired test has $n - 1$ df instead of roughly $n + m - 2$. Almost always a bargain.

9 · Correlation

Whether two variables move together, and what that still cannot tell you.

Does advertising move sales?

A third business. `branches` holds 120 branches of a Mexican retail chain, one row per branch, with last month's figures:

- `ad_spend` — local advertising spend, in thousands of pesos
- `sales` — sales, in thousands of pesos
- `staff` — number of employees

Head office wants to know whether branches that advertise more actually sell more — and neither column splits into groups. Both are numbers.

So this is not a difference in means. The question is whether the two move together.

Hypothesis testing for correlation

Null hypothesis: $H_0 : \rho = \rho_0$, almost always $\rho_0 = 0$ — no linear relationship in the population.

Assume normally distributed variables in independent pairs $\{(a_1, s_1), (a_2, s_2), \dots\}$.

Test statistic and its distribution under the null:

$$T_{test} = \frac{\hat{\rho} - \rho_0}{\sqrt{\frac{1-\hat{\rho}^2}{n-2}}} \sim t_{n-2}$$

your H_A	reject H_0 when
$\rho \neq \rho_0$	$\lvert T_{test} \rvert > t_{n-2,\, 1-\alpha/2}$
$\rho > \rho_0$	$T_{test} > t_{n-2,\, 1-\alpha}$
$\rho < \rho_0$	$T_{test} < t_{n-2,\, \alpha}$

Look before you test

The test assumes the pairs are independent and the relationship is linear. Both are claims about the picture, so draw it first.

Exercise 20 — Draw the relationship

1. Plot `sales` against `ad_spend`.
2. Does the cloud slope? Does the slope look straight, or bent?
3. Guess the correlation before you compute it.

Exercise | Start Over | Show Hint | Run Code

```
1 branches[1:6, ]
```

Exercise 21 — Test it

Head office only cares about one direction: more advertising going with more sales. Test at $\alpha = 0.05$.

1. Write H_0 and H_A.
2. The sample correlation.
3. The test statistic, using $t = \hat{\rho}\sqrt{\frac{n-2}{1-\hat{\rho}^2}}$.
4. The one-sided cut-off, your decision, and the p-value.
5. Then: $\hat{\rho}^2$. How much of the variation in sales does advertising explain?

Exercise | Start Over | Show Hint | Run Code

```
1 a <- branches$ad_spend
2 s <- branches$sales
3 n <- 120
```

What a correlation test cannot do

- It measures linear movement only. A strong bent relationship can give $\hat{\rho}$ near 0.
- It says nothing about cause.
- Significance is about $\rho \neq 0$, not about ρ being large. With $n = 120$ even a correlation of 0.18 would reject.

Which is the whole lesson: a test tells you a relationship is there, and $\hat{\rho}^2$ tells you whether it is worth anything.

Exam spotlight — Fall 2025 Midterm 1

Real Midterm 1 questions on the Households & Environment Survey ($n = 13{,}255$). We solve these in class on `Data/households_environment.rda` (object `exam_data`).

→ More exam practice at the end of this deck

Exam question — Q9 & Q10

Q9. A report claims the true mean monthly electricity bill is 200. Your sample mean is 206. Why can't we immediately conclude the report is wrong?

Q10. State H_0 and H_A for testing whether the true mean monthly electricity bill differs from 200.

Exam question — Q12

Q12. Using the electricity-bill data ($n = 13{,}255$), compute the test statistic for $H_0 : \mu = 200$ vs $H_A : \mu \neq 200$, and give your decision.

Do it here, exam-style: statistic first, then the decision at two significance levels.

Exercise | Start Over | Show Hint | Run Code

```
1 bill <- exam_data$electricity_bill_monthly
2 n    <- 13255
```

Practice now: the one-sample z/t problems in the Week 6 (Testing I) set at the end of this deck.

Exam question — Fall 2025 Midterm 1

Q16. Design an A/B test to measure whether a change causes an increase in an outcome: define the treatment and control groups, the randomization, the outcome, and the two-sample test you would run.

We frame this in class using `Data/households_environment.rda` (object `exam_data`):

- Randomize units into treatment vs control → the groups are, on average, identical except for the treatment.
- Collect the outcome, then run a two-sample difference-in-means test ($H_0 : \mu_T - \mu_C = 0$ against $H_A : \mu_T - \mu_C > 0$, since the question is whether the change causes an increase).

> Practice now: the two-sample and A/B-design problems in the Week (Testing II) set.

→ More exam practice at the end of this deck

Exercise 23 — The whole week, one question

You advise the Airbnb host portfolio. The owner asks:

> *"We are considering paying for a deep-clean service on every listing. Is it worth it — and how sure are you?"*

Answer with a number, a picture, and one sentence. Then add the sentence the owner will not like.

Exercise | Start Over | Show Hint | Run Code

```
clean <- Sample_list$price[Sample_list$clean == TRUE]
dirty <- Sample_list$price[Sample_list$clean == FALSE]

# 1. A number: how much more do clean listings earn, and how uncertain is that figure?

# 2. A picture: something the owner can read in three seconds.

# 3. One sentence for the owner.

# 4. One sentence on what this evidence CANNOT tell them.
```

What you must be able to do this week

State	Write H_0 and H_1 for a claim about a mean, a difference, or a correlation.
Compute	Build the statistic by hand: estimate, claim, standard error, degrees of freedom.
Decide	Compare with the cut-off from `qnorm`/`qt`, or compute the p-value with `pnorm`/`pt`.
Choose	Say which of the four tests a business question calls for, and why.
Pair	Recognise when two columns are the same units measured twice, and test the differences.
Report	Give the number, the decision, and one sentence the manager can act on.

Practice for this week

Weekly exercise set: `W06_exercises.pdf`, with solutions in `W06_solutions.pdf`.

You can already solve exercises 1–20 and 22–26.

Not in these slides: exercise 21, A/B testing (optional, not on the exam).

Suggested before the next class, closest to the exam:

Exercise	Topic	How
7	Courier delivery times: test for one mean	R as a calculator
11	A food-spending claim on a small sample, ENIGH	R with the data
17	Output at the office and at home: two means	R as a calculator
19	Output before and after training: paired data	R as a calculator
24	Food spending by gender of the household head, ENIGH	R with the data

Appendix — optional (extra detail)

Derivations, the second worked example, and the material we did not cover in class.

Where exactly does 1.645 come from?

The rejection region for a one-sided test at 5% is a piece of the standard normal — the piece that holds 5% of the area on the right.

It is the value with exactly 5% of the curve above it: `qnorm(0.95)`.

How far is the truth from the claim?

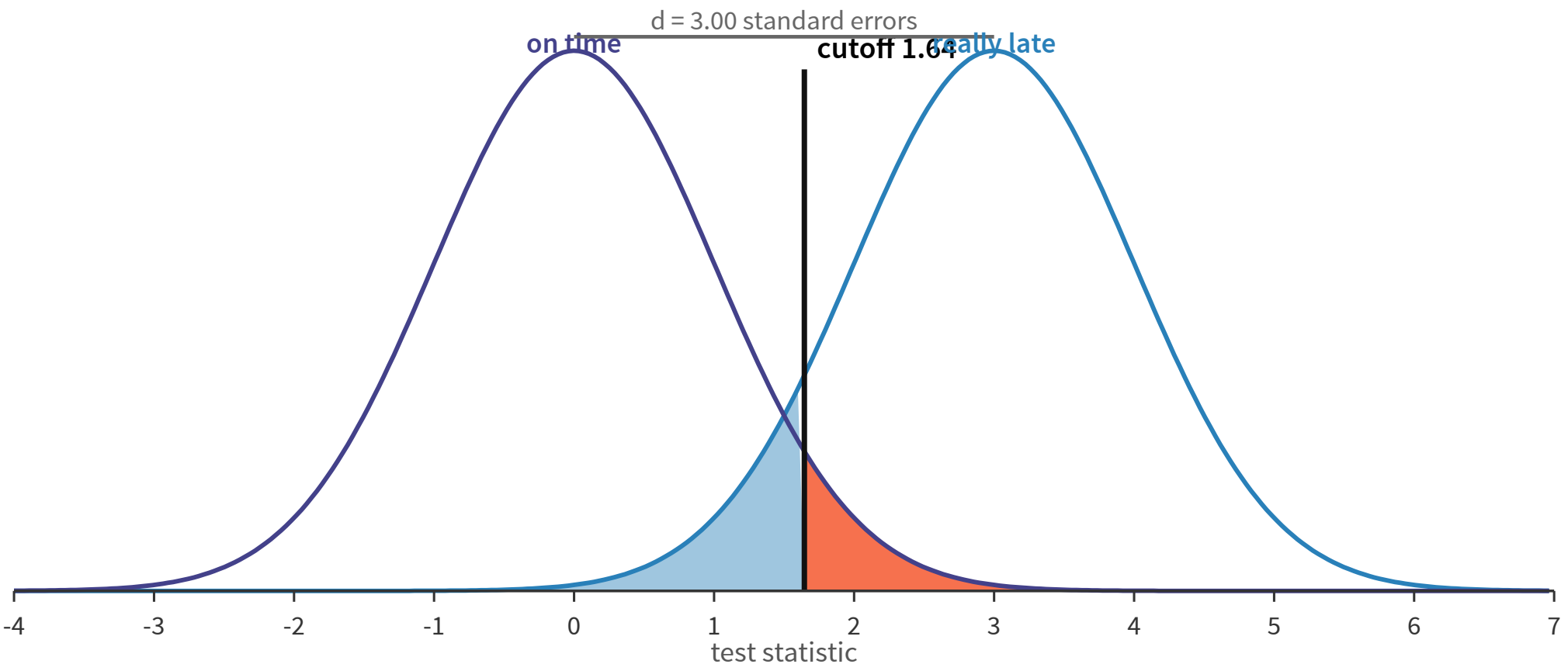
Measure the gap the way we measure everything — in standard errors:

$$d = \frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}} = \frac{110 - 100}{20/\sqrt{36}} = 3$$

significance level α **α = 0.0500** ↔ cutoff **cutoff z = 1.645**

☑ **courier is on time** ☑ **courier is really late** true mean 110 min sample size n 36

reject when the sample mean exceeds 105.5 min **β = 0.088** **power = 0.912**



- Same shape, same standard deviation — the blue curve is just the purple one slid 3 standard errors right.
- The cutoff does not move: it was fixed by α, before anyone knew the truth.
- Blue is β: the courier really is late, but the statistic still lands short of the cutoff, so you do not claim.
- Red is α: the courier is on time, but the statistic overshoots anyway.

β is the area of the shifted curve lying below the cutoff — and the shift is the only thing that changed, so we just read the normal table d further along:

$$\beta = \Phi(z_{1-\alpha} - d) = \Phi(1.645 - 3) = \Phi(-1.355) = 0.088$$

In R that is one line: `pnorm(qnorm(0.95) - 3)`.

So with a courier who really runs 10 minutes late, this test still misses them about 9 times in 100. Go back to the interactive picture, slide the true mean closer to 100, and that number climbs fast — a small delay is much harder to prove than a large one.

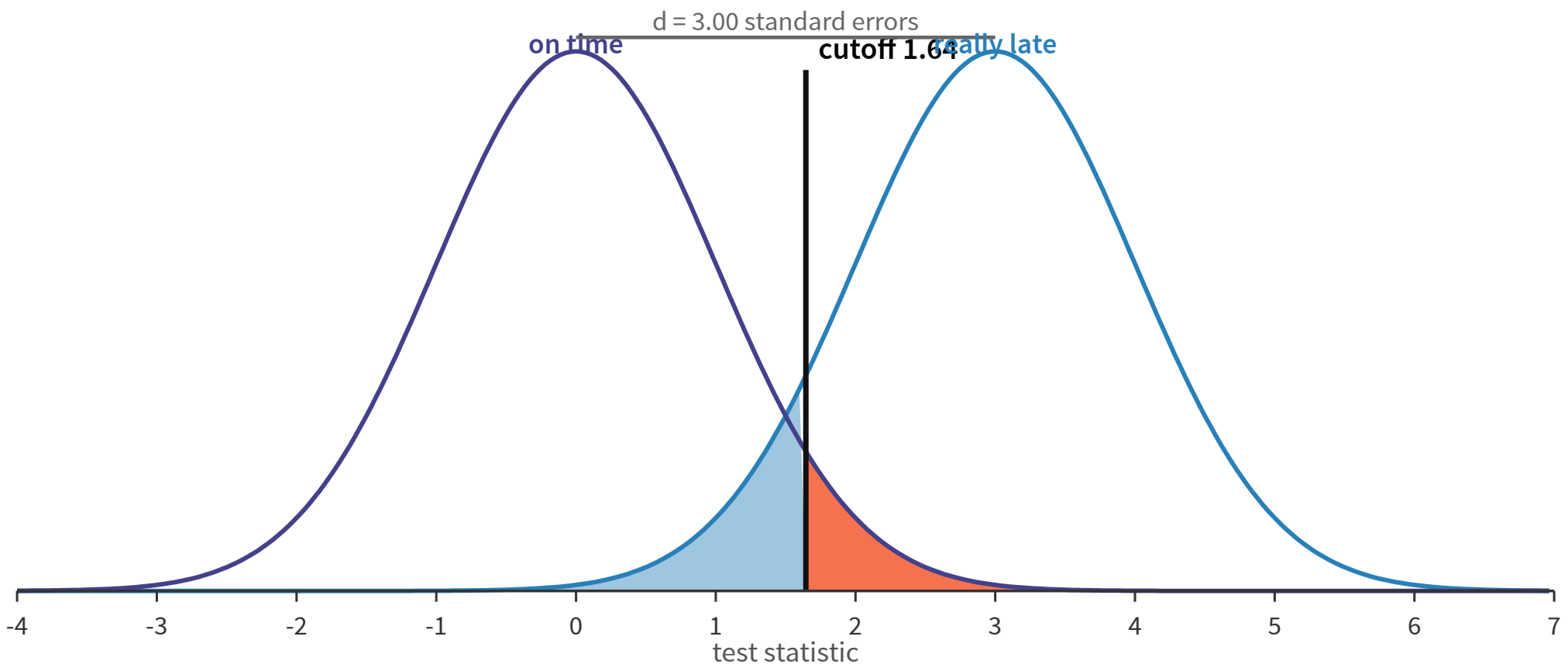
What lowering α costs you

Both curves are on. Now move only the significance level, and watch the two shaded areas at the same time.

significance level α **α = 0.0500** ↔ cutoff **cutoff z = 1.645**

courier is on time **courier is really late** true mean 110 min sample size n 36

reject when the sample mean exceeds 105.5 min **β = 0.088** **power = 0.912**



Set α to 0.10, then 0.05, then 0.01, and read β off each time.

α	cutoff	β	power
0.10	1.282	0.043	0.957
0.05	1.645	0.088	0.912
0.01	2.326	0.250	0.750

The two errors move in opposite directions. Demanding more evidence before you claim makes you wrong less often about an on-time courier, and wrong *more* often about a late one. There is no setting of α that makes both small.

Now change the sample size instead. That is the one lever that shrinks both at once — which is why the honest answer to “how do I get a better test?” is almost always more data.

What makes a test powerful?

Four things could plausibly matter: the sample size n, how far the truth sits from the null, the population standard deviation σ, and the significance level α.

Rather than argue about it — count rejections.

So how do we pick the cutoff?

We have been moving it by hand. In practice you do not choose the cutoff — you choose α, and the cutoff follows.

- α is the error you can control without knowing the truth: it only needs the null world.
- β and power need a specific alternative, which you never actually know.
- So the convention: fix α first — usually 0.05 — and let it set the cutoff.

$$\text{choose } \alpha \longrightarrow \text{cutoff } z_{1-\alpha} \longrightarrow \text{decision}$$

Everything else this week is that same chain, with a different reference distribution depending on the sample size and the alternative.

General Procedure

1. Determine the null and alternative hypotheses

2. Collect your sample of independent observations

3. Choose the appropriate test

4. Calculate the test statistic

5. Compare it to the rejection region

6. Reject or fail to reject the null hypothesis

Let’s talk about appropriate tests!

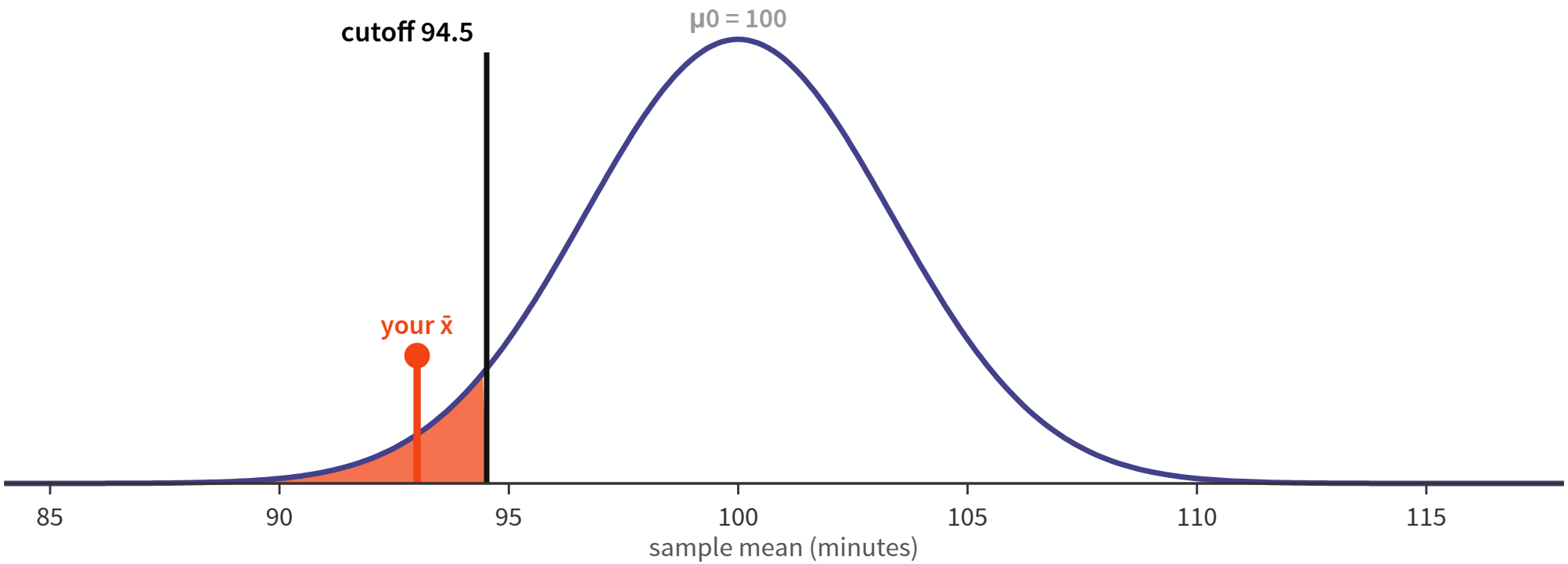
Appendix: reject μ_0, reject everything above it

$H_0 : \mu \geq 100$ against $H_A : \mu < 100$, with $\sigma = 20$ and $n = 36$. The rule is built at the boundary $\mu_0 = 100$: at $\alpha = 0.05$, reject when $\bar{X} < 94.5$.

your sample mean x̄ 93.0 min tested value μ 100.0 min

statistic against μ_0 = 100: -2.10 → reject **statistic against μ = 100.0: -2.10 → reject**

if the true mean were μ, the rule rejects with probability 0.050 (α = 0.050)



- Put your $\bar{X}$ below 94.5, so you reject $\mu_0 = 100$. Then move the tested value μ up. The statistic against μ only gets more negative, so you reject every μ above 100 too.
- The red area is how often the rule rejects if the true mean is μ. It equals α at $\mu = 100$ and shrinks as μ grows.

So the boundary μ_0 is the worst case inside H_0. Control the error there and you control it for every $\mu \geq \mu_0$.

← Back to Case 1

Single Sample - Mean - One sided - Case 2

Hypothesis: $H_0 : \mu = \mu_0$ and $H_A : \mu > \mu_0$

- This includes any null like $\mu \leq \mu_0$. Why?
- If you rejected μ_0 at α, you would reject for sure anything smaller than μ_0.

Rejection Region: for a test of significance level α, reject

- If large sample or known variance from normal

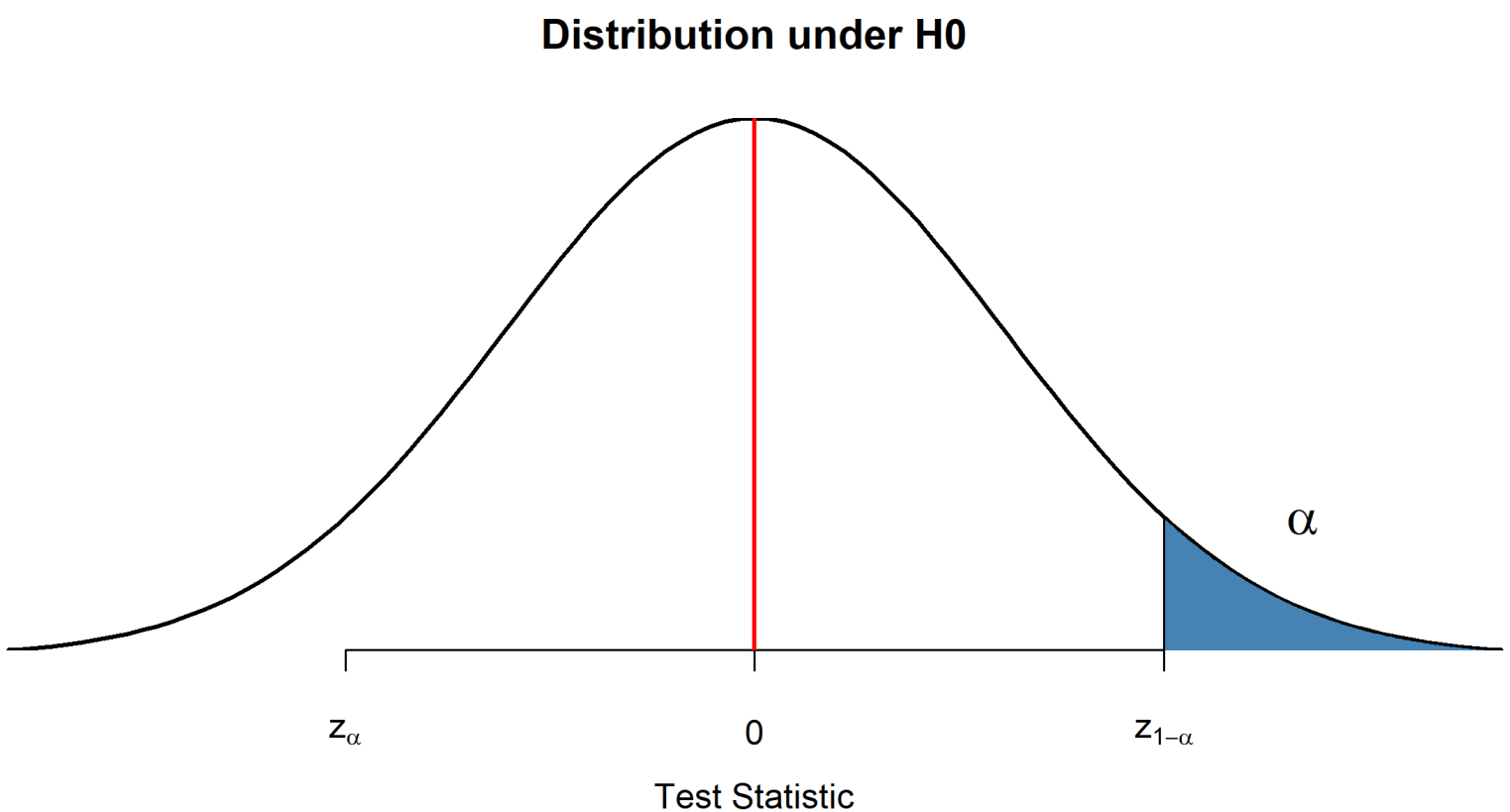
$$\frac{\bar{X} - \mu_0}{SE} > z_{1-\alpha}$$

- If small sample from normal

$$\frac{\bar{X} - \mu_0}{SE} > t_{(n-1),1-\alpha}$$

Where $z_{1-\alpha}$ and $t_{(n-1),1-\alpha}$ are $(1-\alpha)$ quantiles of the standard normal and Student's t with $n-1$ df.

Single Sample - Mean - One sided - Case 2: the picture



Exercise 18 — The whole week, one question

You manage the courier contract. It promises an average delivery time of 100 minutes, and the discount clause is one-sided: you can only claim if they are slower. Your director asks:

> *“Do we invoke the clause, or not?”*

Answer with a number and one sentence.

Exercise | Start Over | Show Hint | Run Code

```
1 mins <- deliveries$minutes
2 n    <- 36
```

The rejection rule

Reject H_0 when the statistic lands in the tail your alternative points to. The statistic is the same number either way — only the distribution you compare it against changes.

your H_A	large sample: $Z_{test} \sim N(0,1)$	small sample: $T_{test} \sim t_v$
$\mu_X - \mu_Y \neq \Delta_0$	$\lvert Z_{test} \rvert > z_{1-\alpha/2}$	$\lvert T_{test} \rvert > t_{v,\,1-\alpha/2}$
$\mu_X - \mu_Y > \Delta_0$	$Z_{test} > z_{1-\alpha}$	$T_{test} > t_{v,\,1-\alpha}$
$\mu_X - \mu_Y < \Delta_0$	$Z_{test} < z_{\alpha}$	$T_{test} < t_{v,\,\alpha}$

At $\alpha = 0.05$ the normal cut-offs are 1.96 two-sided and 1.645 one-sided. The t cut-offs always sit a little further out, and close in on these as v grows.

The two-sided rule splits α into two halves, so each tail gets only $\alpha/2$. That is why its cut-off sits further from zero than the one-sided one.

Where exactly do you cut?

The rule says “reject in the tails”. The tails have to be *somewhere*: two numbers on the horizontal axis, chosen so the shaded area equals α.

For $\alpha = 0.05$, two-sided: what are those two numbers under $N(0, 1)$?

And if the sample is small so the statistic follows $t(v)$ instead — do the cutoffs move in or out?

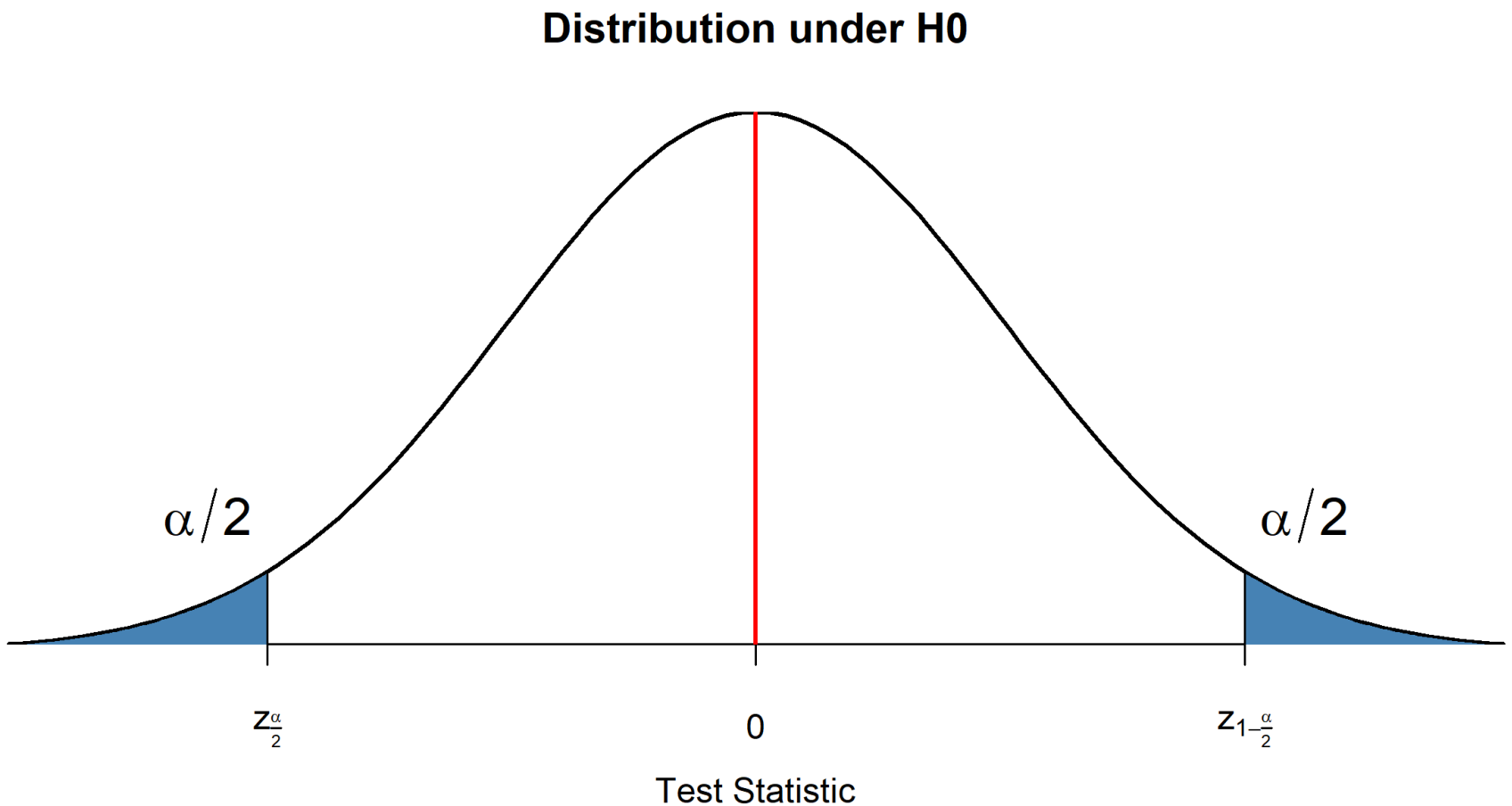
Exercise 1 — Find the cutoffs

1. Compute the two cutoffs that leave $\alpha/2 = 0.025$ in each tail of $N(0, 1)$.
2. Compute the same upper cutoff from Student's t with $v = 5, 20, 100, 1000$ degrees of freedom, and say which way they move as v grows.

Exercise | Start Over | Show Hint | Run Code

```
1 alpha <- 0.05
```

Two-sided rejection region



Two samples — mean — one-sided — case 1

Hypothesis: $H_0 : \mu_X - \mu_Y = \Delta_0$ and $H_A : \mu_X - \mu_Y < \Delta_0$

- This includes any null like $\mu_X - \mu_Y \geq \Delta_0$. Why?
- If you rejected $\mu_X - \mu_Y = \Delta_0$ at α, you would reject for sure anything larger than Δ_0. Why, in a picture →

Rejection Region: for a test of significance level α, reject

- If large sample or known variance from normal

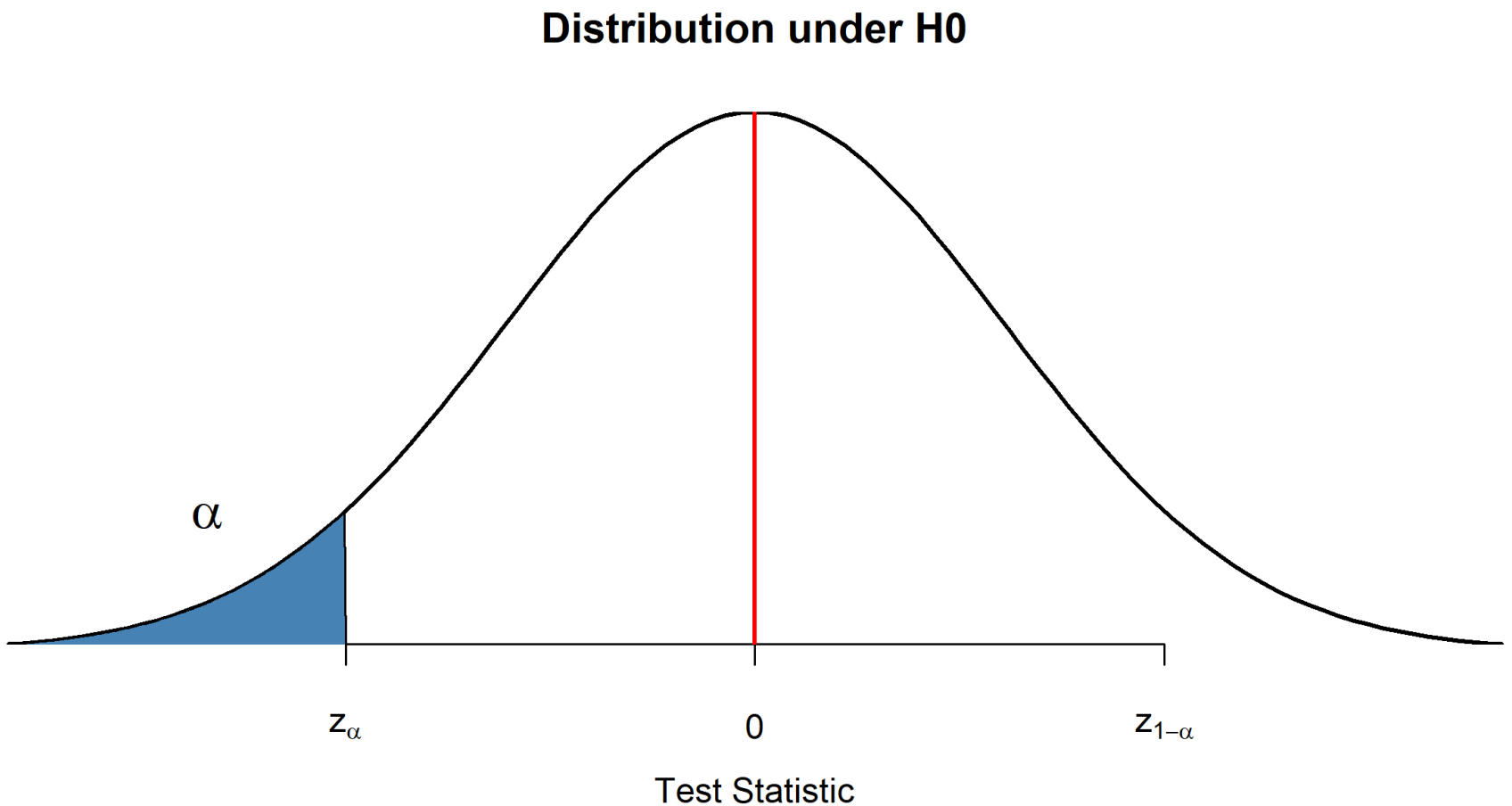
$$\frac{\bar{X} - \bar{Y} - (\Delta_0)}{SE} < z_\alpha$$

- If small sample from normal

$$\frac{\bar{X} - \bar{Y} - (\Delta_0)}{SE} < t_{v,\alpha}$$

Where z_α and $t_{v,\alpha}$ are α quantiles of the standard normal and Student's t with v df.

One-sided rejection region — case 1



Two samples — mean — one-sided — case 2

Hypothesis: $H_0 : \mu_X - \mu_Y = \Delta_0$ and $H_A : \mu_X - \mu_Y > \Delta_0$

- This includes any null like $\mu_X - \mu_Y \leq \Delta_0$. Why?
- If you rejected $\mu_X - \mu_Y = \Delta_0$ at α, you would reject for sure anything smaller than Δ_0. Why, in a picture →

Rejection Region: for a test of significance level α, reject

- If large sample or known variance with normal

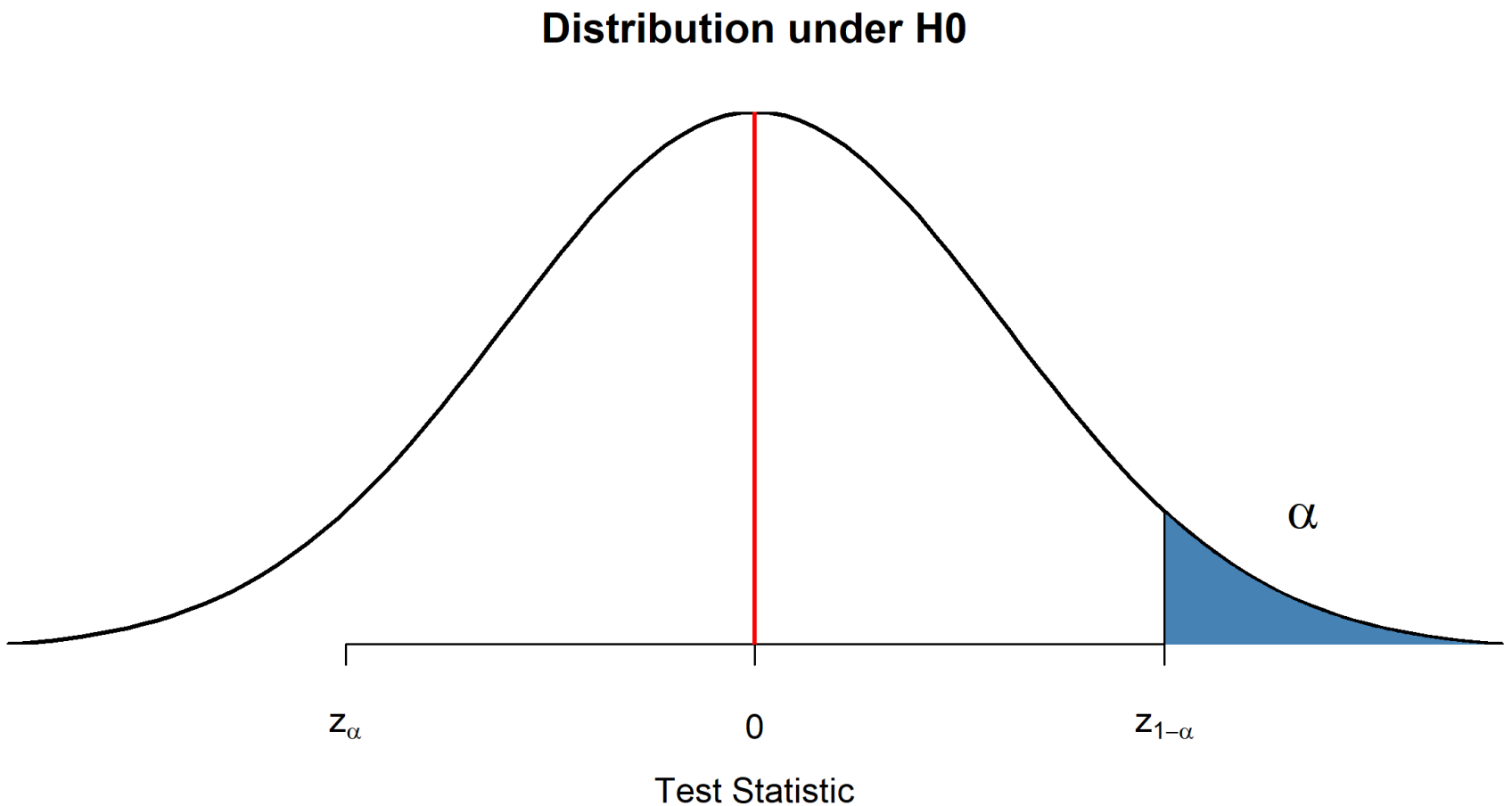
$$\frac{\bar{X} - \bar{Y} - (\Delta_0)}{SE} > z_{1-\alpha}$$

- If small sample from normal

$$\frac{\bar{X} - \bar{Y} - (\Delta_0)}{SE} > t_{v,1-\alpha}$$

Where $z_{1-\alpha}$ and $t_{v,1-\alpha}$ are $(1-\alpha)$ quantiles of the standard normal and Student's t with v df.

One-sided rejection region — case 2



A second business: the delivery app

A food-delivery app in Mexico City pulled a small audit of recent orders. `orders` has 25 rows:

- `channel` — `"app"` or `"web"`, how the order was placed
- `basket` — what the customer spent, in pesos

Small, because pulling and checking each order by hand is expensive. Keep that in mind — it will matter.

Exercise | Start Over | Show Solution | Run Code

```
1 orders
2
3 table(orders$channel)
```

Should we push customers to the app?

Building and maintaining the app costs money. It is only worth it if app customers actually spend more.

- $H_0 : \mu_{app} - \mu_{web} = 0$ — the channel makes no difference
- $H_A : \mu_{app} - \mu_{web} > 0$ — app orders are bigger

One-sided: if app baskets turned out smaller, we would drop the idea just as we would if they were equal. Only one direction leads anywhere.

Exercise 6 — Is the app basket bigger?

Work the whole test at $\alpha = 0.05$.

1. The two group means and the gap.
2. The standard error of the gap.
3. The test statistic.
4. Degrees of freedom by the shortcut, and the one-sided cut-off.
5. Your decision, and the one-sided p-value.
6. Now redo step 4 and 5 pretending the samples were large, so the cut-off comes from the normal instead. Does anything change?

Exercise | Start Over | Show Hint | Run Code

```
app <- orders$basket[orders$channel == "app"]
web <- orders$basket[orders$channel == "web"]
n <- 14
m <- 11
```

Paired data — rejection region

Rejection Region: for a test of significance level α:

- If large sample or known variance with normal distribution:
- for $H_A : \bar{d} \neq \Delta_0$, reject if:

$$\frac{\bar{d} - \Delta_0}{SE} < z_{\frac{\alpha}{2}} \quad or \quad \frac{\bar{d} - \Delta_0}{SE} > z_{1-\frac{\alpha}{2}}$$

- for $H_A : \bar{d} > \Delta_0$, reject if:

$$\frac{\bar{d} - \Delta_0}{SE} > z_{1-\alpha}$$

- for $H_A : \bar{d} < \Delta_0$, reject if:

$$\frac{\bar{d} - \Delta_0}{SE} < z_{\alpha}$$

- If small sample from normal, replace critical values with the t-distribution with n-1 degrees of freedom.

Exercise 22 — Correlation with n = 27

Only two numbers are given, and that is enough. Test $H_0 : \rho = 0$ against $H_A : \rho \neq 0$.

1. Compute the test statistic and its degrees of freedom.
2. Compute the two-sided p-value and the two critical values at $\alpha = 0.05$.
3. State the decision.
4. Then ask a second question: with $n = 27$, how large would $|\hat{\rho}|$ have to be to reach significance at all? Solve it by trying a grid of values.

Exercise | Start Over | Show Hint | Run Code

```r
rho <- -0.45
n   <- 27

# your code here
```

Appendix: from the statistic back to the sample mean

- Suppose I reject if $z > 1.645$, with $\sigma_x = 20$. How large must the mean be to reject?

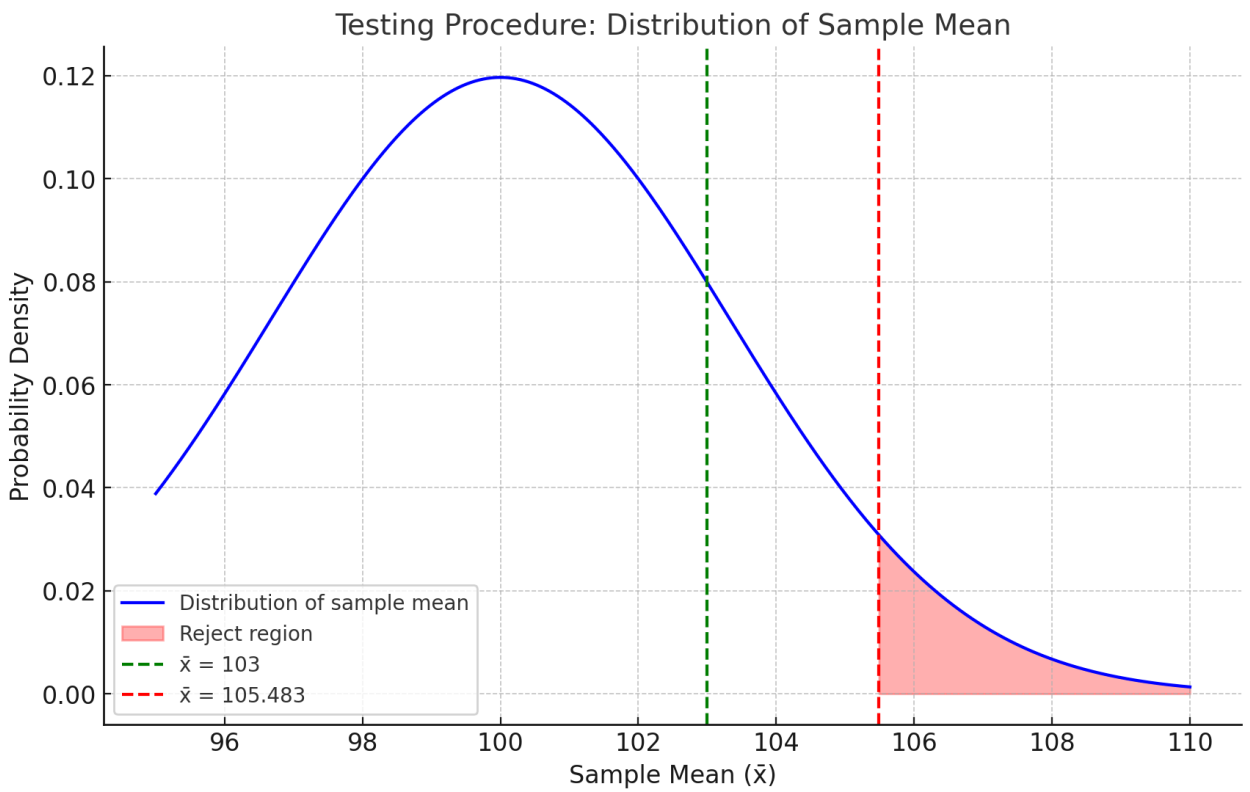
Reject if:

- $\frac{\bar{x}-100}{20/\sqrt{36}} > 1.645$
- Or equivalently if $\bar{x} > 105.483$
- Those two formulations are equivalent.

- Why don't we reject even if $\bar{x} = 103$?

Appendix: why 103 is not enough

- Because there is randomness in the sample:
- Even if $\mu = 100$, we could get a sample with $\bar{x} > 100$.



- Why $\bar{x} > 105.483$ though? Formally later, but it's very unlikely to get a sample mean that large if $\mu = 100$.

Appendix: the Type 2 error, derived step by step

- What's the probability our sample mean is below the critical value, if the courier's true mean is above 100?
- If I take 10000 samples from a distribution with true mean > 100, what share of sample means fall below the rejection region?
- In our example:

$$\beta = P(\text{Type 2 error}) = \underbrace{P_{H_A}(z < 1.645)}_{\text{Pr. if alt is true}} = P_{H_A}(\frac{\bar{x} - 100}{20/\sqrt{36}} < 1.645)$$

- Problem: how is $\frac{\bar{x}-100}{20/\sqrt{36}}$ distributed if the alternative is true?
- CLT does not apply to $\frac{\bar{x}-100}{20/\sqrt{36}}$ because 100 is not the true mean under the alternative.
- We need to specify the true mean. Say $\mu_1 = 110$ — the courier really averages 110 minutes. Then CLT applies to $\frac{\bar{x}-110}{20/\sqrt{36}}$.

Appendix: the Type 2 error, derived (cont.)

- Rewrite so we have $\frac{\bar{x}-110}{20/\sqrt{36}}$ on the left-hand side.

$$\beta = P_{H_A}(\frac{\bar{x}-100}{20/\sqrt{36}} < 1.645) = P_{H_A}(\bar{X} < 105.483) = P\left(\frac{\bar{X}-110}{20/\sqrt{36}} < \frac{105.483-110}{20/\sqrt{36}}\right)$$

- Now this becomes:

$$P\left(z < \frac{105.483-110}{20/\sqrt{36}}\right) = P(z < -1.3551)$$

- Where z is normally distributed thanks to CLT.

Appendix: the general Type 2 formula

- More generally for this type of test:

$$\beta = P_{H_A}(\bar{X} < \mu_0 + z_{1-\alpha}\frac{\sigma_x}{\sqrt{n}}) = P\left(\frac{\bar{X} - \mu_1}{\sigma_x/\sqrt{n}} < \frac{\mu_0 - \mu_1}{\sigma_x/\sqrt{n}} + z_{1-\alpha}\right)$$

- Depends on the true μ and where the rejection region is.

Appendix: what α costs you in β, in R

Courier setting: $\mu_0 = 100, \sigma = 20, n = 36$, and suppose the truth is $\mu_1 = 110$ minutes.

For $\alpha = 0.10, 0.05$ and 0.01, compute and tabulate:

1. the critical value $z_{1-\alpha}$,
2. the sample mean at which you would start rejecting, $\mu_0 + z_{1-\alpha}\sigma/\sqrt{n}$,
3. β from the formula on the previous slide,
4. $1 - \beta$.

Then say which way each column moves as α falls.

Exercise | Start Over | Show Hint | Run Code

```r
mu0 <- 100; mu_true <- 110; sigma <- 20; n <- 36
alphas <- c(0.10, 0.05, 0.01)
se <- sigma / sqrt(n)

# 1. the three critical values

# 2. the sample mean at which you start rejecting

# 3. beta

# 4. one minus beta
```

Appendix: the trade-off, stated

If I change the critical value $z_{1-\alpha}$, it has the opposite effect on α and β. Choosing α is choosing which of the two errors you would rather make.

Interactive visualization: https://shiny.rit.albany.edu/stat/betaprob/

Appendix: power, for four possible truths

Courier test again: $H_0 : \mu = 100$ against $\mu > 100, \sigma = 20, \alpha = 0.05$.

Power is $1 - \Phi\left(z_{1-\alpha} - \frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}}\right)$. Suppose the courier's true mean is 102, 105, 110 or 115 minutes.

1. Compute the power at each of those four truths with $n = 36$.
2. Compute the same four with $n = 100$.
3. Say which of the two — a bigger effect or a bigger sample — moves power more here.

Exercise | Start Over | Show Hint | Run Code

```r
mu0 <- 100; sigma <- 20
zc  <- qnorm(0.95)
mus <- c(102, 105, 110, 115)

# 1. power at n = 36

# 2. power at n = 100
```

Exercise 10, confirmed by counting

The formula is doing nothing more than counting rejections. This code counts them directly, for a truth of 105 at both sample sizes. Press Run and compare with the numbers you just computed.

Exercise | Start Over | Show Solution | Run Code

```
set.seed(3)
z36 <- replicate(2000, (mean(rnorm(36, 105, 20)) - 100) / (20 / sqrt(36)))
mean(z36 > qnorm(0.95))

z100 <- replicate(2000, (mean(rnorm(100, 105, 20)) - 100) / (20 / sqrt(100)))
mean(z100 > qnorm(0.95))
```

Appendix: draw the two-sided rejection region

Exercise 5 put all 5% on the right. A two-sided test splits it. Redraw the picture for $H_A : \mu \neq \mu_0$ at $\alpha = 0.05$:

1. Both critical values.
2. Both shaded tails.
3. Check that the two shaded areas together really are 5%.

Then, in one line: which of the two cut-offs, 1.645 or the one you just got, is further from zero — and why?

Exercise | Start Over | Show Hint | Run Code

```r
# 1. The two cut-offs:
zc <- ______
zc

# 3. The two shaded areas, added together:
______

# The picture (written for you):
curve(dnorm(x), xlim = c(-3, 3), lwd = 2, xlab = "Test statistic", ylab = "")
lo <- seq(-3, zc[1], 0.01); hi <- seq(zc[2], 3, 0.01)
polygon(c(-3, lo, zc[1]), c(0, dnorm(lo), 0), col = "steelblue")
polygon(c(zc[2], hi, 3),  c(0, dnorm(hi), 0), col = "steelblue")
abline(v = zc, col = "red", lwd = 2)
```

Appendix: test and confidence interval, two answers to one question

Last week you built a 95% confidence interval for a mean. This week you test whether the mean equals 200. Two different procedures, same data.

Can they contradict each other?

Appendix: the test and the interval agree

Build both intervals and both decisions yourself.

1. The 95% interval. Does it contain 100? Does the test reject at $\alpha = 0.05$?
2. Break it: now the 99% interval. Does it still exclude 100? Does the decision still agree with the interval?

Exercise | Start Over | Show Hint | Run Code

```
1 mins <- deliveries$minutes
2 n    <- 36
```

Appendix: link between tests and confidence intervals

Test for μ with $H_0 : \mu = 4$ and $H_A : \mu \neq 4$.

- Population standard deviation known, $\sigma = 3$. Sample of 49 with mean $\bar{x} = 3$.
- Test statistic: $z_{test} = \frac{3-4}{\sigma/\sqrt{49}} = \frac{3-4}{3/7} = -2.34$
- Critical values at $\alpha = 0.05$ are -1.96 and 1.96, hence we reject.
- The 95% confidence interval for the mean:

$$\left(3 - 1.96\frac{\sigma}{\sqrt{49}},\ 3 + 1.96\frac{\sigma}{\sqrt{49}}\right) = (2.16,\ 3.84)$$

- It does not contain the null hypothesis.

Appendix: the link, algebraically

- More generally: the $1-\alpha$ CI doesn't contain the null $\iff$ the null would be rejected with a test of significance α.

https://kzaremba.shinyapps.io/Hypothesis_Confidence/

- Mathematically: let μ_0 be the null. In a two-sided test we reject at α if

$$\left|\frac{\bar{X}-\mu_0}{s/\sqrt{n}}\right| > z_{1-\frac{\alpha}{2}}$$

  - Which can be rewritten as:

$$\mu_0 < \bar{X} - z_{1-\frac{\alpha}{2}}\frac{s}{\sqrt{n}} \qquad or \qquad \mu_0 > \bar{X} + z_{1-\frac{\alpha}{2}}\frac{s}{\sqrt{n}}$$

Appendix: moving all of α into one tail

The two-sided picture splits α into two halves. A one-sided alternative puts all of it on one side.

Take your two-sided code and change only where the shading goes.

Appendix: one tail instead of two

Starting from your Exercise 12 code, at $\alpha = 0.05$:

1. Draw the left-tail picture ($H_A : \mu < \mu_0$) and give its critical value.
2. Draw the right-tail picture ($H_A : \mu > \mu_0$) and give its critical value.
3. Put the three critical values side by side (two-sided, left, right) and say which alternative makes a statistic of -1.8 significant, and which does not.

Exercise | Start Over | Show Hint | Run Code

```r
# 1. The left-tail cut-off:
lo <- ______

# 2. The right-tail cut-off:
hi <- ______

# 3. The three cut-offs side by side: two-sided upper, left, right
lo
hi
______

# The two pictures (written for you):
par(mfrow = c(1, 2))
curve(dnorm(x), xlim = c(-3, 3), lwd = 2, main = "HA: mu < mu0", xlab = "z", ylab = "")
s1 <- seq(-3, lo, 0.01)
```

```r
polygon(c(-3, s1, lo), c(0, dnorm(s1), 0), col = "steelblue")
abline(v = lo, col = "red", lwd = 2)

curve(dnorm(x), xlim = c(-3, 3), lwd = 2, main = "HA: mu > mu0", xlab = "z", ylab = "")
s2 <- seq(hi, 3, 0.01)
polygon(c(hi, s2, 3), c(0, dnorm(s2), 0), col = "steelblue")
abline(v = hi, col = "red", lwd = 2)
par(mfrow = c(1, 1))
```

Appendix: Type 1 error, counted in R

The code runs 5,000 audits in a world where the null is true: the courier really does average $\mu = 100$ minutes, $\sigma = 20$, $n = 36$. The courier is on time every time, and every audit is done correctly. It is written for you — press Run, then answer from the output.

1. What share of these 5,000 audits reject a null that is true?
2. How does that share change at the 10% and the 1% critical values?
3. What does that tell you about what α actually measures?

Exercise | Start Over | Show Solution | Run Code

```r
set.seed(11)
z <- replicate(5000, (mean(rnorm(36, mean = 100, sd = 20)) - 100) / (20 / sqrt(36)))

mean(z > qnorm(0.90))
mean(z > qnorm(0.95))
mean(z > qnorm(0.99))

hist(z, breaks = 50, col = "grey80",
     main = "5,000 studies where H0 is TRUE", xlab = "Test statistic")
abline(v = qnorm(0.95), col = "red", lwd = 3)
```

Appendix: what α really means

α is not a description of one study. It is the long-run rate at which a correct null gets rejected: run the same correct audit 5,000 times at $\alpha = 0.05$ and roughly 250 of them report a “finding”.

That is the price of the rejection region — and lowering it has a cost of its own.

Appendix: what a Type 1 error means here

In our courier test ($H_0 : \mu = 100$), what would a Type 1 error mean?

- Claiming the discount when the courier's true mean is really 100 minutes
- Failing to claim the discount when the courier is really slower than 100
- Collecting a sample that is too small
- Choosing the wrong reference distribution

Claiming the discount when the courier's true mean is really 100 minutes.

Type 1 error = rejecting a true H_0 (a "false positive"). Its probability is the significance level α. Failing to reject a false H_0 is a Type 2 error.

Appendix: the critical values in R

Take the picture you just drew and change one thing: the significance level.

1. Compute the one-sided critical value for $\alpha = 0.10, 0.05, 0.01$ and 0.
2. Say what happens to the width of the rejection region as α falls, and what the $\alpha = 0$ case means in practice.

Exercise | Start Over | Show Hint | Run Code

```r
alphas <- c(0.10, 0.05, 0.01, 0)

# The four critical values:
zc <- ______
zc

# The picture (written for you):
curve(dnorm(x), xlim = c(-3, 4), lwd = 2, xlab = "Test statistic", ylab = "")
abline(v = zc[1:3], col = c("darkgreen", "red", "purple"), lwd = 2)
```

Appendix: draw the one-sided rejection region

The drawing code is written for you: it uses whatever `zc` you define.

1. Fill in the cut-off that leaves exactly 5% of the area to its right.
2. Check that the area to the right of it really is 5%.

Exercise | Start Over | Show Hint | Run Code

```r
# 1. The cut-off leaving 5% of the standard normal above it:
zc <- ______
zc

# 2. The area to the right of it:
______

# The picture (written for you):
curve(dnorm(x), xlim = c(-3, 3), lwd = 2, xlab = "Test statistic", ylab = "")
xs <- seq(zc, 3, 0.01)
polygon(c(zc, xs, 3), c(0, dnorm(xs), 0), col = "steelblue")
abline(v = zc, col = "red", lwd = 2)
```

Appendix: the same benchmark, built in R

The interactive version is quicker to watch, but here is the same thing in R: 3,000 samples drawn from a world where the courier is meeting the contract, 36 deliveries each. Press Run, then answer from the output.

1. Where are the 3,000 statistics centred, and what is their standard deviation?
2. The red curve is the standard normal. How well does it fit the histogram?
3. The orange line is our observed statistic. What share of the simulated worlds produce a statistic at least that far from 0?

Exercise | Start Over | Show Solution | Run Code

```r
mins  <- deliveries$minutes
z_obs <- (mean(mins) - 100) / (sd(mins) / sqrt(36))

set.seed(6)
null_z <- replicate(3000, {
  sim <- rnorm(36, mean = 100, sd = sd(mins))
  (mean(sim) - 100) / (sd(sim) / sqrt(36))
})

hist(null_z, breaks = 40, freq = FALSE, col = "steelblue",
     main = "The statistic in 3,000 worlds where the courier keeps the promise",
     xlab = "Test statistic")
curve(dnorm(x), add = TRUE, col = "red", lwd = 2)
abline(v = z_obs, col = "orange", lwd = 3)
```

```r

mean(null_z)
sd(null_z)
mean(abs(null_z) >= abs(z_obs))
```

Appendix: confidence intervals from last week

`exam_data` is the Households & Environment Survey — one row per household, 14,505 rows, with `household_size` giving the number of people living in the home.

1. Compute the mean and the standard deviation of `exam_data$household_size`, then the standard error of the mean (the sample size is 14505).
2. Build the 95% confidence interval for the mean household size.
3. Rebuild the same interval at 99% and say what changed.

Exercise | Start Over | Show Hint | Run Code

```
1 # task 1: xbar, s, and the standard error
2
3 # task 2: the 95% interval
4
5 # task 3: the same interval at 99%
```

Appendix: confidence interval for a difference

Following this idea, we can construct a confidence interval for the difference:

$$P\left(z_{\alpha/2} < \frac{\bar{X} - \bar{Y} - (\mu_{X,0} - \mu_{Y,0})}{\sqrt{s_X^2/n + s_Y^2/m}} < z_{1-\alpha/2}\right) = 1 - \alpha$$

$$CI_{1-\alpha} = \left(\bar{X} - \bar{Y} \pm z_{1-\alpha/2}\sqrt{s_X^2/n + s_Y^2/m}\right)$$

Appendix: the interval for a difference, by hand

Build the 95% confidence interval for the difference yourself, from the gap and the standard error you already have.

Exercise | Start Over | Show Hint | Run Code

```
gas  <- exam_data$gasoline_spending_weekly
cool <- exam_data$uses_electricity_cooling
yes  <- gas[cool == TRUE]
no   <- gas[cool == FALSE]

gap <- mean(yes, na.rm = TRUE) - mean(no, na.rm = TRUE)
SE  <- sqrt(var(yes, na.rm = TRUE)/7746 + var(no, na.rm = TRUE)/6759)
```

Does the 95% CI for the difference contain 0?

No — the interval runs from about 89 to 115, so 0 is nowhere near it.

Appendix: decision at α = 0.05

Look at the two numbers you just produced in Exercise 3 — nothing else.

At $\alpha = 0.05$, what is the correct decision for H_0: the two group means are equal, against H_A: they differ?

Reject H_0 — the p-value came out far below 0.05, and the CI excludes 0.

The rule: reject H_0 when $p < \alpha$ — equivalently, when the CI excludes Δ_0. (Rejecting only "if we first assume equal variances" is wrong — Welch's test needs no such assumption.)

Appendix: the whole α in one tail

Two-sided at $\alpha = 0.05$ splits the 5% into two tails of 2.5%. One-sided puts all 5% into a single tail.

Write down your prediction before running anything: does the one-sided cutoff sit further from zero or closer to zero than 1.96?

And what does that do to the p-value of a statistic that points in the alternative's direction?

Appendix: one-sided cutoffs

Your prediction: the one-sided cutoff is _____ than 1.96.

Now produce the evidence.

1. Compute the two-sided cutoff and the one-sided cutoff at $\alpha = 0.05$.
2. Take a test statistic of -2.10. Compute its two-sided p-value and its one-sided ("less") p-value under $N(0, 1)$.
3. State the exact arithmetic relationship between those two p-values.

Exercise | Start Over | Show Hint | Run Code

```
1 # Your code here
```

Appendix: the two-sample test by hand

Use only the six summary numbers from the table above — no raw data — to test $H_0 : \mu_{clean} - \mu_{dirty} = 0$ against a two-sided alternative.

1. The standard error of the difference.
2. The test statistic.
3. The degrees of freedom, both ways: the full Welch formula from the "Side note" slide, and the shortcut $v = \min(n - 1, m - 1)$.
4. The two-sided p-value and the 95% CI.

Exercise | Start Over | Show Hint | Run Code

```
1  n_c <- 100; mean_c <- 1245.43; sd_c <- 961.90   # clean
2  n_d <- 100; mean_d <-  868.70; sd_d <- 693.31   # dirty
3
4  A <- sd_c^2/n_c
5  B <- sd_d^2/n_d
```

Appendix: when is a test paired?

Which of these is the condition that makes a test paired?

- the two samples have the same number of observations
- each unit contributes one observation to each of the two measurements
- the two samples come from the same population
- the outcome is continuous

Each unit contributes one observation to each measurement — the same employee, store or neighborhood measured twice. Then the two columns line up row by row, and you test the within-unit differences d_i.

Equal sample sizes are a *consequence* of pairing, not the cause: two independent samples can happen to have $n = m$ and still be unpaired. If you can shuffle one column independently of the other without losing information, the data are not paired.

Appendix: does IQ decrease with age?

A psychologist thinks age influences IQ. They take a random sample of 100 people aged 40. For each person we know their IQ at age 16 and now — two measurements on the same person.

On average, IQ at the young age was 8 points higher than at age 40, with a standard deviation of that difference of 7 points.

Using $\alpha = 0.01$, test the hypothesis that IQ decreases with age.

Appendix: the IQ test, from summaries

You are given only n, $\bar{d}$ and s_d — which is all a paired test needs.

1. Write H_0 and H_A for "IQ decreases with age", with $d_i = \text{IQ}_{16} - \text{IQ}_{40}$.
2. Compute the standard error and the test statistic.
3. Compute the p-value and the critical value at $\alpha = 0.01$.
4. State the decision, and say why this is a paired problem and not a two-sample one.

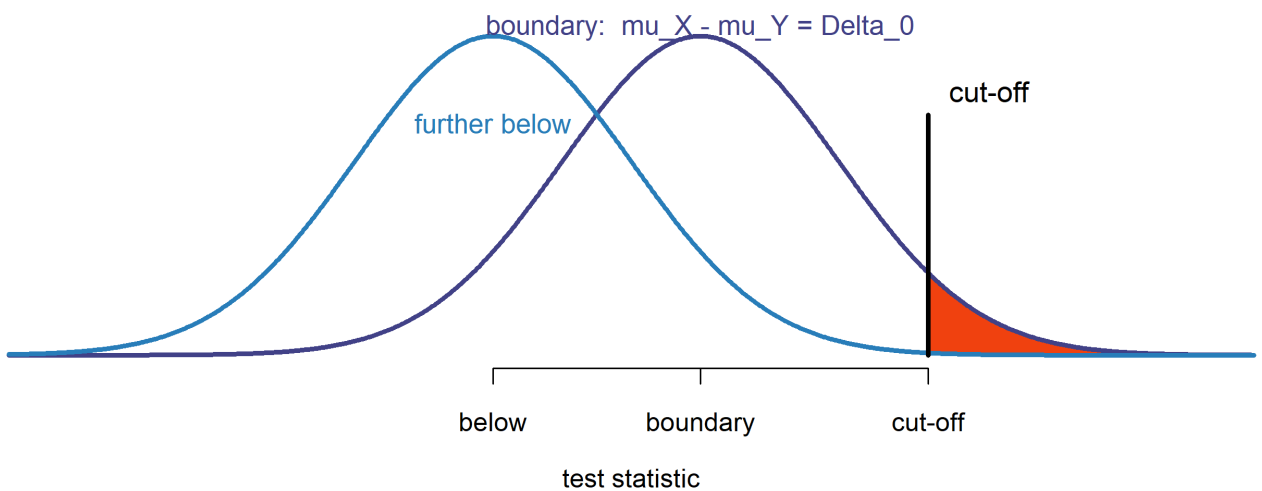
Exercise | Start Over | Show Hint | Run Code

```r
n    <- 100    # people
dbar <- 8      # mean of (IQ at 16 - IQ at 40)
s_d  <- 7      # sd of those differences

# your code here
```

Appendix: why the boundary is enough

We write $H_0 : \mu_X - \mu_Y = \Delta_0$, but we really mean "$\Delta_0$ or less" (for an upper-tail test). Testing only the boundary is enough. Here is why.



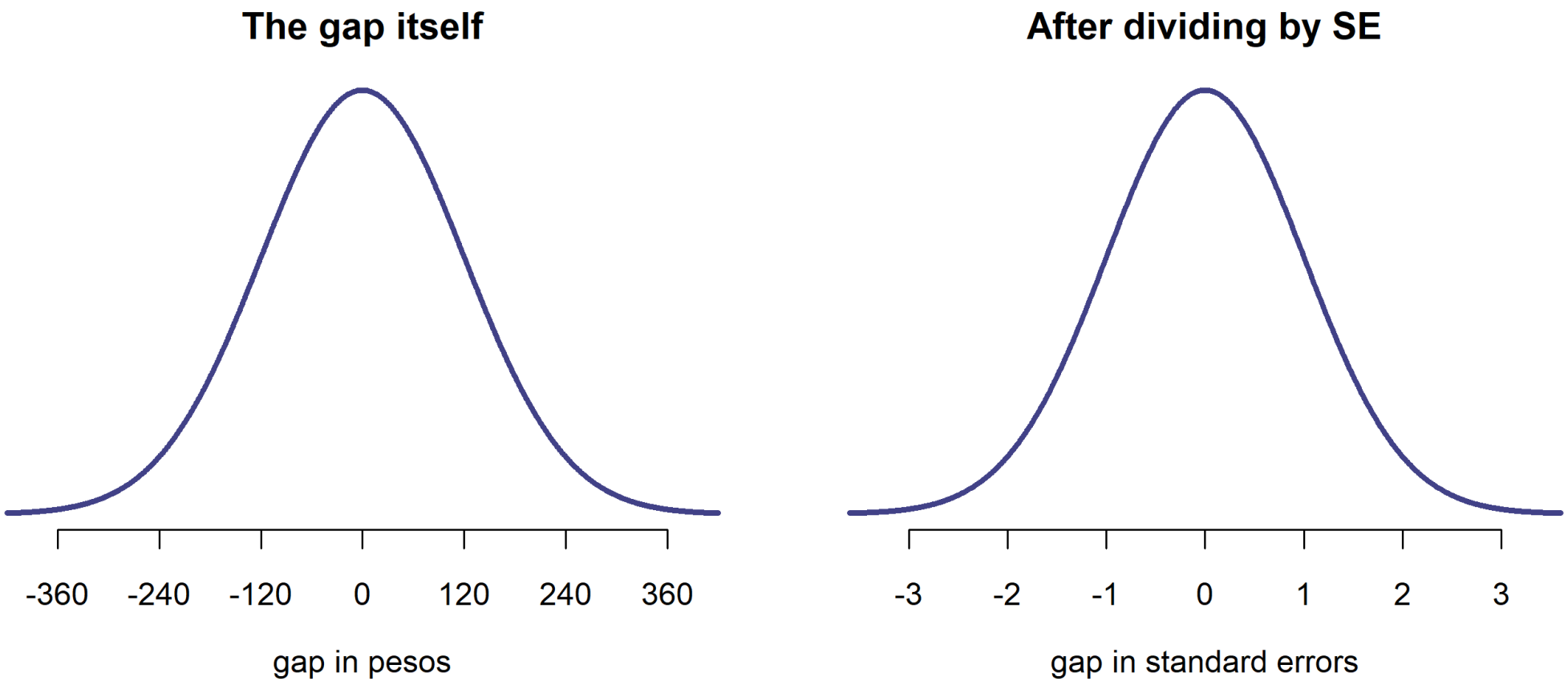
- The purple curve is the boundary case: the gap is exactly Δ_0. The red area is how often it rejects — that is α.
- The blue curve is a null further below the boundary. It sits to the left, so *less* of it lands past the cut-off. It rejects less often than α.

So the boundary is the worst case: any null further from the alternative is even safer. Control the error rate there and you have controlled it everywhere in H_0.

That is why we can write the null as a single equals sign and forget the rest.

← Back to the one-sided test

Appendix: same gap, two rulers



- The same distribution, drawn twice. Only the ruler changed.
- On the left the axis is pesos, and the width depends on the data.
- On the right the axis is standard errors, and under H_0 the width is always 1.

That is why cut-offs like 1.96 can be memorised: they live on the right-hand ruler, which never changes.

Not covered this term: A/B testing

The slides below are kept for reference. They are not examinable.

Appendix: A/B testing — the business question





- Airbnb wants to know: *Do professional photos increase host revenue?*
- Historical data: listings with better photos show higher revenues.
- But… Correlation ≠ Causation. Better photos may be linked to other factors: more organized hosts, higher-quality listings, better service.
- A simple comparison only tells us about association, not causality.

The business decision

- Should Airbnb invest in professional photography for hosts?
- If photos aren't the true driver → the investment wastes money. We need a method to isolate the effect of photos.

Enter A/B Testing

Definition:

- A method to measure *causal* effects.
- Widely used in modern business, especially tech.

Examples:

- Netflix: which thumbnail gets more clicks?
- Amazon: does a new checkout button increase sales?
- Instagram: do longer videos increase engagement?

The setup

We divide hosts into two groups:

1. Control group: no intervention (existing photos).
2. Treatment group: Airbnb provides professional photography.

The KEY: groups are assigned randomly.

Exercise 10 — What makes an A/B test valid

Four candidate answers. Which single feature is what lets an A/B test measure a causal effect?

- a large enough sample in each arm
- running the test long enough to see the effect
- assigning units to treatment and control at random
- measuring an outcome that management cares about

Random assignment. It is the only one of the four that makes the two groups comparable on *everything you did not measure*.

Sample size and duration buy precision — a narrower CI around whatever you estimate. Randomization buys validity — the guarantee that what you estimate is the treatment effect and not a difference between the kinds of units in each arm.

Measuring the effect

- After randomization: the only difference is the photos.
- Provide photography to the treatment group
- Wait to see potential effects
- Collect outcome data (revenue, reviews, bookings)
- Run a simple two-sample test between treatment and control
- If the treatment group earns more revenue, we know it was caused by the professional photos

This is the causal effect.

- A/B testing is used extensively in modern business, especially in tech.
- We have all been subjects of these experiments, often without knowing.

Exercise 11 — A/B test as a proportion test

An A/B test on a checkout button. The control group saw the old button; the treatment group saw the new one. Conversions (purchases) are:

Group	Conversions	Visitors
Control	120	2000
Treatment	155	2000

The outcome here is binary, so the two-sample test is a test of proportions. Compute the two conversion rates and the lift.

Exercise | Start Over | Show Hint | Run Code

```
x_control <- 120;  n_control <- 2000
x_treat   <- 155;  n_treat   <- 2000

# 1. the two conversion rates

# 2. the lift, in percentage points
```

Exercise 12 — Reading the proportion test

Reuse the A/B test from Exercise 11. Test $H_0 : \pi_T = \pi_C$ against $H_A : \pi_T \neq \pi_C$: build the z-statistic by hand from the pooled proportion, find its two-sided p-value, and reach a decision at $\alpha = 0.05$.

Exercise | Start Over | Show Hint | Run Code

```
x1 <- 120; n1 <- 2000    # control
x2 <- 155; n2 <- 2000    # treatment

# 1. the pooled proportion under H0

# 2. the standard error under H0

# 3. the z-statistic and its two-sided p-value
```

Does randomization really balance the groups?

The claim is strong: split units at random and the two arms end up identical on *everything* — household size, income, region, and every variable you never even recorded.

That is a claim you can check. Take units that were never treated at all, split them at random, and compare their characteristics across the fake “arms”.

If the claim is right, what should the comparison show?

Exercise 13 — Check the randomization

Use `exam_data` (14,505 households). Nobody is treated — the code only *pretends* to assign them, then checks whether the two fake arms look alike. It is written for you — press Run, then answer:

1. Do the two arms match on `household_size`, `electricity_bill_monthly` and `num_vehicles`? What about the shares of `locality_size`?
2. The balance t-test's p-value: what should it look like here, and what does it look like?
3. The last block repeats the whole thing ten times. Does any single balance test ever come out "significant"?

Exercise | Start Over | Show Solution | Run Code

```r
set.seed(1)
arm <- sample(c("A", "B"), 14505, replace = TRUE)

size <- exam_data$household_size
bill <- exam_data$electricity_bill_monthly
cars <- exam_data$num_vehicles

mean(size[arm == "A"]);  mean(size[arm == "B"])
mean(bill[arm == "A"], na.rm = TRUE);  mean(bill[arm == "B"], na.rm = TRUE)
mean(cars[arm == "A"]);  mean(cars[arm == "B"])

round(prop.table(table(arm, exam_data$locality_size), 1), 3)

bA <- bill[arm == "A"]
```

```r
bB <- bill[arm == "B"]
z  <- (mean(bA, na.rm = TRUE) - mean(bB, na.rm = TRUE)) /
      sqrt(var(bA, na.rm = TRUE)/7250 + var(bB, na.rm = TRUE)/7255)
2 * (1 - pnorm(abs(z)))

# and again, ten more times:
replicate(10, {
  a  <- sample(c("A", "B"), 14505, replace = TRUE)
  xA <- bill[a == "A"]; xB <- bill[a == "B"]
  zz <- (mean(xA, na.rm = TRUE) - mean(xB, na.rm = TRUE)) /
        sqrt(var(xA, na.rm = TRUE)/7250 + var(xB, na.rm = TRUE)/7255)
  2 * (1 - pnorm(abs(zz)))
})
```

Why randomization works

- Both arms are two samples from the same population — so every expectation, observed or not, is the same in both.
- Therefore any post-treatment difference in the outcome can only come from the treatment.
- **This is what gives us causal inference.**

Without randomization the groups differ in ways you cannot list, let alone control for.

If the button does nothing, how often will you still "find" an effect?

Your team ships an A/B test on a button that — unknown to everyone — changes nothing at all. True conversion rate: 6% in both arms.

You run the test on 2,000 visitors per arm and look at the p-value.

Write down your guess: out of 1,000 such tests on a *useless* button, how many come back significant at $\alpha = 0.05$?

And what shape does the histogram of those 1,000 p-values have?

Exercise 14 — An A/B test with no effect

Your guess: _____ out of 1,000.

The simulation is written for you. One “experiment” draws conversions for both arms from the *same* 6% rate, then tests. Press Run, then answer: what share came back significant, and what shape is the histogram?

Exercise | Start Over | Show Solution | Run Code

```r
one_test <- function(n = 2000, p_control = 0.06, p_treat = 0.06) {
  xC   <- rbinom(1, n, p_control)
  xT   <- rbinom(1, n, p_treat)
  phat <- (xC + xT) / (2 * n)
  z    <- (xT/n - xC/n) / sqrt(phat * (1 - phat) * 2 / n)
  2 * pnorm(-abs(z))
}

set.seed(3)
pvals <- replicate(1000, one_test())

mean(pvals < 0.05)

hist(pvals, breaks = 20, col = "steelblue",
     main = "1,000 A/B tests on a button that does nothing",
     xlab = "p-value")
abline(v = 0.05, col = "red", lwd = 3)
```

Two ways to turn 5% into much more

The 5% above assumed you ran one test, on one metric, and looked once.

Real teams do neither:

- They watch the dashboard daily and stop as soon as it goes green.
- They track conversion, revenue, retention, clicks, time-on-page, … and report whichever moves.

Both feel harmless. Predict how far each one pushes the 5% before you measure it.

🖌️ Exercise 15 — Break it: peek, and test many metrics

Still a button that does nothing. Reuse `one_test()` from Exercise 14.

a. Peeking. Instead of testing once at 2,000 visitors per arm, test after every 200 visitors and stop the moment any p-value dips below 0.05. Out of 1,000 such experiments, how often do you stop and declare a winner?

b. Many metrics. Back to testing once, but on 10 independent metrics, reporting a win if *any* of them is significant. How often does that happen? Compute the exact probability too.

Both simulations are written for you. Press Run, then compare the two numbers with the 5% you measured in Exercise 14 — and with your own predictions.

Exercise | Start Over | Show Solution | Run Code

```r
one_test <- function(n = 2000, p_control = 0.06, p_treat = 0.06) {
  xC   <- rbinom(1, n, p_control)
  xT   <- rbinom(1, n, p_treat)
  phat <- (xC + xT) / (2 * n)
  z    <- (xT/n - xC/n) / sqrt(phat * (1 - phat) * 2 / n)
  2 * pnorm(-abs(z))
}

set.seed(11)

# (a) peeking every 200 visitors, stopping at the first p < 0.05
```

```r
peek <- function(n = 2000, step = 200, p = 0.06) {
  cC <- cumsum(rbinom(n, 1, p))
  cT <- cumsum(rbinom(n, 1, p))
  ks <- seq(step, n, by = step)
  pv <- sapply(ks, function(k) {
    phat <- (cC[k] + cT[k]) / (2 * k)
    z    <- (cT[k]/k - cC[k]/k) / sqrt(phat * (1 - phat) * 2 / k)
    2 * pnorm(-abs(z))
  })
  any(pv < 0.05, na.rm = TRUE)
}
mean(replicate(500, peek()))

# (b) ten metrics at once
mean(replicate(500, any(replicate(10, one_test()) < 0.05)))

# (b) the exact probability
1 - 0.95^10
```

Sizing the experiment: professional photos

- Airbnb wants to test if professional photos increase weekly revenue.
- Design:
  - Control: no photos (baseline $\mu_0 = 2000$)
  - Treatment: professional photos (suspected $\mu_a = 2200$)
  - $\sigma = 600$, $n = 100$ per group, $\alpha = 0.01$
- Steps:
  1. State hypotheses: $H_0 : \mu_T - \mu_C = 0$ vs $H_A : \mu_T - \mu_C > 0$
  2. Find the critical value
  3. Compute power $= 1 - \beta$
  4. Find minimum n for 80% power

Exercise 10b — Run the A/B power calculation

Work steps 2–4 in R. The statistic is a difference of two means, so its standard error combines both groups: $SE = \sqrt{\sigma^2/n + \sigma^2/n}$.

1. The critical value, expressed as a revenue difference in pesos.
2. The power of the planned test at $\mu_a - \mu_0 = 200$.
3. The n per group that would buy 80% power, from $n = 2\sigma^2 \left(\frac{z_{1-\alpha} + z_{1-\beta}}{\mu_a - \mu_0} \right)^2$.

Exercise | Start Over | Show Hint | Run Code

```r
mu0 <- 2000; mua <- 2200; sigma <- 600; n <- 100; alpha <- 0.01

# (1) SE of the difference, then the critical difference in pesos

# (2) power at a true gap of 200

# (3) n per group for 80% power
```

How many visitors do you need?

Now flip it around. Suppose the new button genuinely lifts conversion from 6% to 7% — a real effect, worth money.

Run the test on 2,000 visitors per arm and you might still miss it. The probability of *catching* a real effect is the test's power.

Guess the power at $n = 2{,}000$ per arm. Then guess how large n has to be for a 90% chance of catching a 1-percentage-point lift.

Exercise 16 — Sample size by simulation

Your guesses: power at $n = 2{,}000$ is _____; the n needed for 90% power is _____.

The code sets the two arms to genuinely different rates — $p_C = 0.06$ vs $p_T = 0.07$ — and counts how often the test catches the difference, at five sample sizes. Press Run, then answer:

1. What is the power at $n = 2{,}000$ per arm?
2. Roughly what n buys you 90% power?
3. What does that mean for how long the experiment must run if the site gets 5,000 visitors a day?

Exercise | Start Over | Show Solution | Run Code

```r
one_test <- function(n, p_control, p_treat) {
  xC   <- rbinom(1, n, p_control)
  xT   <- rbinom(1, n, p_treat)
  phat <- (xC + xT) / (2 * n)
  z    <- (xT/n - xC/n) / sqrt(phat * (1 - phat) * 2 / n)
  2 * pnorm(-abs(z))
}

set.seed(5)
ns    <- c(1000, 2000, 5000, 10000, 20000)
power <- sapply(ns, function(n) mean(replicate(300, one_test(n, 0.06, 0.07)) < 0.05))
round(power, 3)

plot(ns, power, type = "b", pch = 19, col = "steelblue", lwd = 2,
```

```
     xlab = "Visitors per arm", ylab = "Power",
     main = "Chance of detecting a 6% -> 7% lift")
abline(h = 0.9, col = "red", lty = 2, lwd = 2)
abline(h = 0.05, col = "grey50", lty = 3)
```

Appendix: the one-sample test from last week

`Sample_list` holds the 200 sampled Mexico City listings, with `price` the nightly price in pesos. A pricing report claims the true mean nightly price is 1000.

1. Compute the mean and the standard deviation of `Sample_list$price`, then the standard error of the mean (the sample size is 200).
2. Build the one-sample test statistic for $H_0 : \mu = 1000$ against $H_A : \mu \neq 1000$.
3. Turn it into a two-sided p-value with `pt()` at 199 degrees of freedom.

Exercise | Start Over | Show Hint | Run Code

```
1 # task 1: xbar, s, and the standard error
2
3 # task 2: the test statistic against 1000
4
5 # task 3: the two-sided p-value
```

Exam practice

These problems mirror the weekly exercise set and the exam. Work each one in the cell, then press Solution.

📝 Exam practice 1 — Predicted income: t-statistic and p-value

A random sample of 15 Mexican university students reports predicted first-job income: $\bar{x} = 46{,}000$ pesos, $s = 9{,}000$ pesos. [I4]

a. Compute the t-statistic for $H_0 : \mu_X = 48{,}000$ against $H_A : \mu_X \neq 48{,}000$. (c) Compute the p-value (two-sided).

Exercise | Start Over | Show Hint | Run Code

```
1 xbar <- 46000; s <- 9000; n <- 15; mu0 <- 48000
2 # (a) the standardized distance between the sample mean and the null value
3
4 # (c) the two-sided p-value from the right reference distribution
```

b. Do you reject H_0 at 5% in a two-sided test?

No — fail to reject. The critical value is $t_{0.025,\,14} = 2.145$ (`qt(0.975, 14)`), and $|t| = 0.861 < 2.145$. Equivalently, the p-value $0.404 > 0.05$.

Exam practice 2 — Predicted income: a larger sample

Same setting as the previous slide ($\bar{x} = 46{,}000$, $s = 9{,}000$, $H_0 : \mu_X = 48{,}000$ against $H_A : \mu_X \neq 48{,}000$). [I4]

d. If $n = 1{,}000$ instead of $n = 15$, would you reject at 5%?

Exercise | Start Over | Show Hint | Run Code

```
1  xbar <- 46000; s <- 9000; mu0 <- 48000
2  # Redo Exam practice 1 with n = 1000 and compare the two verdicts
```

e. For (d), is the normality assumption necessary?

No. With $n = 1{,}000$ the Central Limit Theorem makes the sampling distribution of $\bar{X}$ approximately normal regardless of the underlying distribution of income. (With $n = 15$ we needed normality to justify the t distribution.)

Exam practice 3 — Candy taste test

250 ITAM students take a blind taste test of green vs. red candies: 140 prefer green, 110 prefer red. [I5]

a. Compute the asymptotic 95% CI for π. (c) Compute the p-value for the test of equal preference, $H_0 : \pi = 0.5$ against $H_A : \pi \neq 0.5$.

Exercise | Start Over | Show Hint | Run Code

```
n <- 250; green <- 140
# (a) the sample proportion, its standard error, and the 95% interval

# (c) the test statistic for equal preference, then its two-sided p-value
```

b. State the null hypothesis for equal preference.

$H_0 : \pi = 0.5$ — students are equally likely to prefer green or red. The alternative is two-sided, $H_A : \pi \neq 0.5$.

Exam practice 4 — Smoking rate

The smoking rate among Mexican adults aged 25–44 is 10.5%. A public health researcher shows an informational video to a random sample. [I6]

a. State the null and alternative hypotheses for testing whether the video decreases smoking prevalence.

$H_0 : \pi \geq 0.105$ vs $H_1 : \pi < 0.105$ — a one-sided (lower-tail) test: the skeptical position is that the video does not reduce smoking.

b. If the follow-up survey finds 8.5% smokers, how large must the sample be to reject the null at 5%?

Exercise | Start Over | Show Hint | Run Code

```
pi0 <- 0.105; pihat <- 0.085
# The rejection condition is an inequality in n -- rearrange it and solve for n
```

Exam practice 5 — Gubernatorial election

In a gubernatorial election in Mexico, older voters (65+) support candidate A with probability 68%; younger voters ($<$65) support candidate B with probability 55%. Suppose 400 older and 1,100 younger voters turn out, A and B are the only options, and everyone votes. [I7]

a. What is the asymptotic distribution of the sample proportion of votes for candidate A?

b. What is the approximate 95% probability interval for A's sample vote share?

c. What is the approximate probability that candidate A wins?

Exercise | Start Over | Show Hint | Run Code

```
n_old <- 400; n_young <- 1100
# (a) the overall probability of an A vote, and the SE of A's vote share

# (b) an interval around it holding about 95%

# (c) the chance A's share lands above one half
```

📝 Exam practice 6 — Stock returns: is the mean return zero?

You are considering investing in a Mexican company listed on the *Bolsa Mexicana de Valores*. Its annual returns over the last 10 years are: $0.03,\ -0.12,\ 0.15,\ 0.04,\ -0.14,\ -0.06,\ 0.00,\ 0.08,\ 0.20,\ 0.07$. Assume returns are i.i.d. normal. [I3]

a. What is the t-statistic for $H_0 : \mu = 0$ against $H_A : \mu \neq 0$? Do you reject at 5%?

b. What is the p-value?

Exercise | Start Over | Show Hint | Run Code

```r
returns <- c(0.03, -0.12, 0.15, 0.04, -0.14, -0.06, 0.00, 0.08, 0.20, 0.07)

# (a) the t-statistic, and the critical value you compare it to

# (b) the p-value
```

c. How would the t-statistic for $H_0 : \mu = -0.01$ against $H_A : \mu \neq -0.01$ compare to (a)?

Larger in absolute value: $t = \uparrow rac0.025 - (-0.01)0.0346 = 1.012$, because -0.01 is farther from $\square arx = 0.025$ than 0 is.

d. How would the p-value compare to (b)?

Exam practice 1 — Domestic vs international scores

Use the `metricsgrades` dataset (68 students in a graduate econometrics course). The key variable is `total`, a composite course score; the dataset also indicates whether each student is `domestic` or `international`. [I21]

Group summaries computed from the data:

Group	n	Mean	Variance
Domestic	32	77.719	50.002
International	36	76.236	67.332

a. Provide an asymptotic 95% confidence interval for the difference in mean total scores between domestic and international students.

b. Conduct a z-test of equal means, $H_0 : \mu_D = \mu_I$ against $H_A : \mu_D \neq \mu_I$. What is the p-value?

Exercise | Start Over | Show Hint | Run Code

```
n_D <- 32;  mean_D <- 77.719;  var_D <- 50.002    # domestic
n_I <- 36;  mean_I <- 76.236;  var_I <- 67.332    # international

# (a) a 95% CI for the difference in mean total scores

# (b) the z-statistic and its two-sided p-value
```

Does the 95% CI contain 0 — and what do you conclude at $\alpha = 0.05$?

Exam practice 2 — Home Depot vs Lowe's returns

Use the `sp500` dataset, which includes monthly returns for several U.S. stocks and the overall market index. Key variables: `HD` (Home Depot returns), `LOW` (Lowe's returns), `IDX` (market index returns). [I22]

Sample summaries ($n = 364$ months): mean HD $= 0.0167$, var HD $= 0.0054$; mean LOW $= 0.0203$, var LOW $= 0.0084$.

a. Compute the z-statistic and p-value for testing $H_0 : \mu_{HD} = \mu_{LOW}$ against $H_A : \mu_{HD} \neq \mu_{LOW}$, and state the decision at $\alpha = 0.05$.

Exercise | Start Over | Show Hint | Run Code

```
n <- 364
mean_HD  <- 0.0167; var_HD  <- 0.0054
mean_LOW <- 0.0203; var_LOW <- 0.0084

# the z-statistic and its two-sided p-value
```

Exam practice 2 (cont.) — testing against a benchmark

b. Suppose the market index mean return is 0.0078. Would it be appropriate to test $H_0 : \mu_{HD} = 0.0078$ against $H_A : \mu_{HD} \neq 0.0078$?

First produce the number the question is really about: run the one-sample z-test of HD against that value, so you can see what it would claim.

Exercise | Start Over | Show Hint | Run Code

```
n <- 364
mean_HD <- 0.0167; var_HD <- 0.0054
benchmark <- 0.0078

# the one-sample z-statistic and its two-sided p-value
```

Now the question itself: is that test appropriate?

It depends on where 0.0078 comes from. If it is an *external benchmark* (a historical average or theoretical prediction), treating it as a fixed constant in a one-sample test is fine.

But if 0.0078 is the *sample* mean of the index computed from the same data — and HD is a component of that index — then $\bar{X}_{HD}$ and the benchmark are dependent, which violates the one-sample test's assumptions. The SE above is then wrong, and the p-value of 0.021 should not be trusted. A paired or regression-based approach would be more appropriate.

Exam practice 3 — Wages and children: correlation test

Use the `cps` dataset (2809 employed individuals). Variables: `wagehr` (hourly wage in dollars), `ownchild` (number of children in the household). Test whether the correlation between wages and number of children is zero. [I23]

State the null and alternative hypotheses.

$$H_0 : \rho = 0 \qquad H_1 : \rho \neq 0$$

On the full data the sample correlation is $r = 0.0151$ over 1839 complete pairs. Rebuild the test statistic and p-value from those two numbers, then state the decision at $\alpha = 0.05$.

Exercise | Start Over | Show Hint | Run Code

```r
r <- 0.0151
n <- 1839      # complete pairs

# the test statistic, its degrees of freedom, and the two-sided p-value
```

At $\alpha = 0.05$, what do you conclude?

Do NOT reject H_0 — $r = 0.015$ is essentially zero and the p-value is 0.518. There is no statistically significant linear relationship between hourly wages and number of own children.